Bayesian Inference

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Everything you learned about statistics so far is **wrong***!

*just kidding. Mostly.

Estimating Probabilities for Rare Events

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What would you do?

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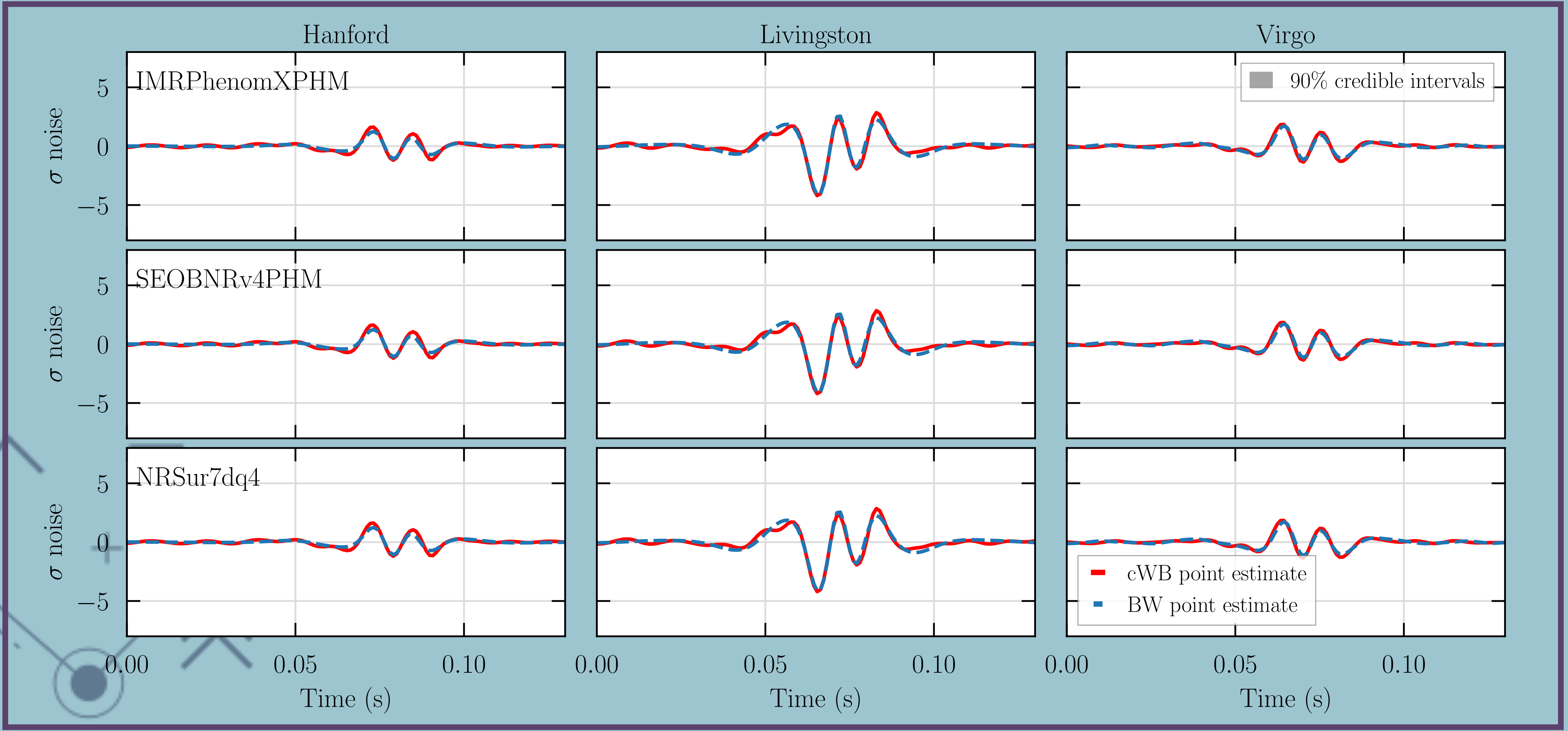
Result: expect 19 “conspicuous weapon accidents” per year. Led to implemetation of major safety features.

A related Astronomy problem:

estimating event rates in astronomical surveys



The LVK Collaboration (2021)

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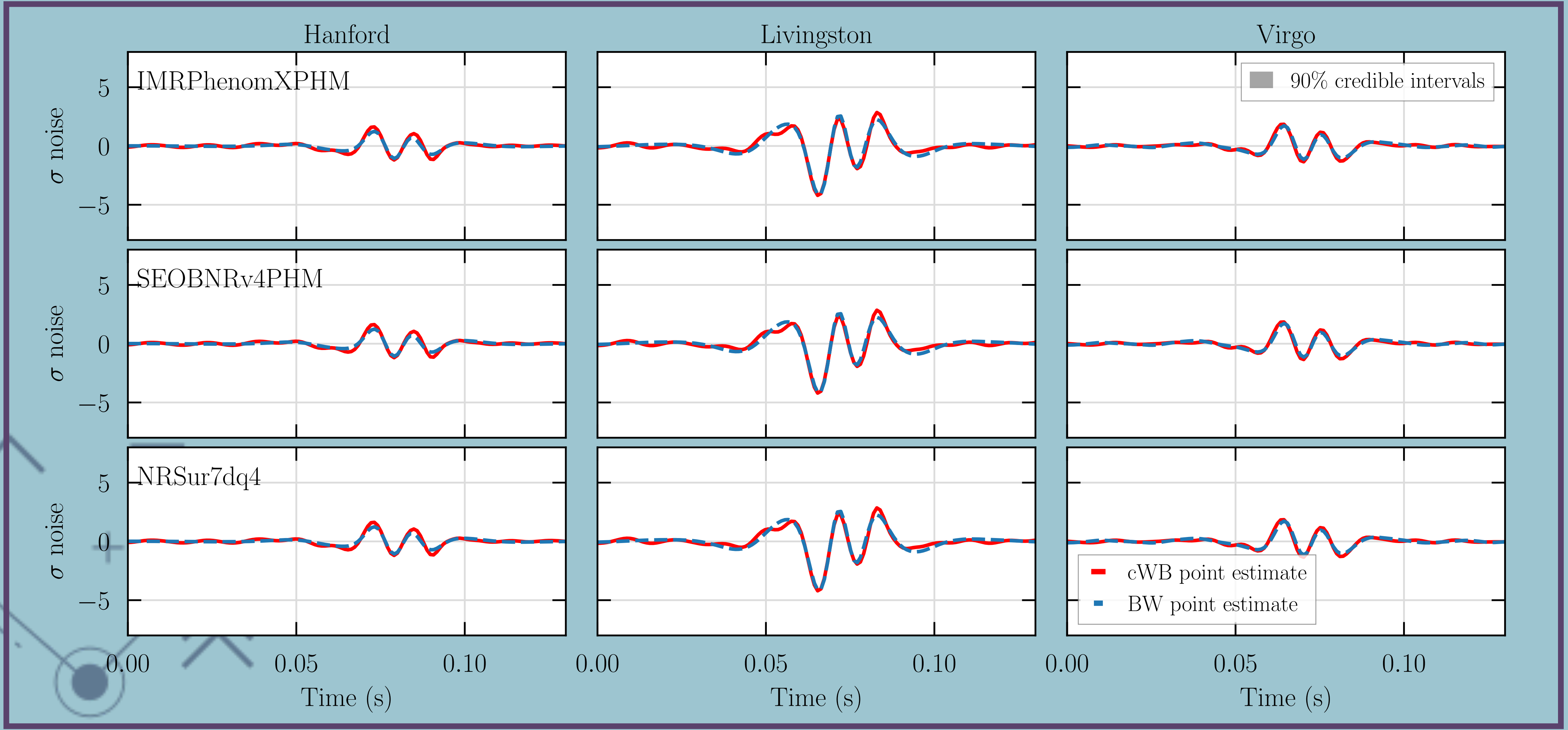
LIGO has found a single GW event from an intermediate-mass black hole.

What is the underlying rate of of IMBH mergers?

What is the predicted number of IMBH GW events in Advanced LIGO?



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- update your probability based on data


Is it a random variable?
https://app.wooclap.com/STATPHYSASTRO25

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IS A RANDOM VARIABLE!

Statistics Battle of the Century

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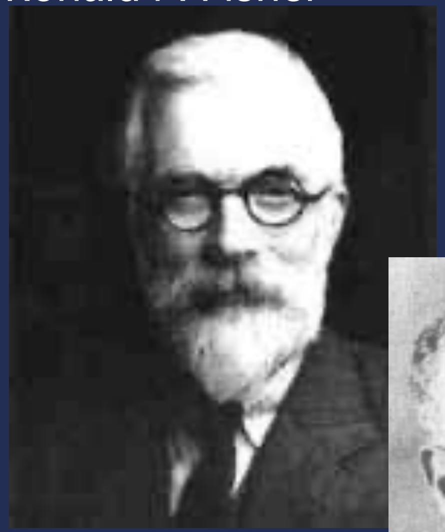
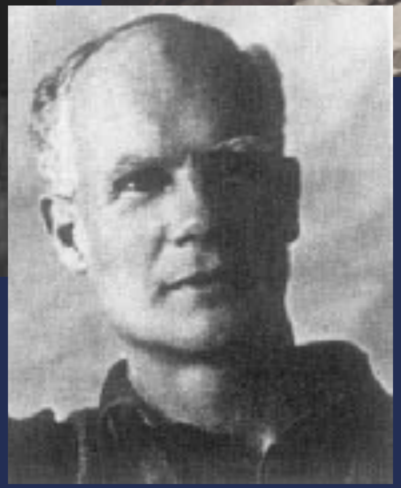
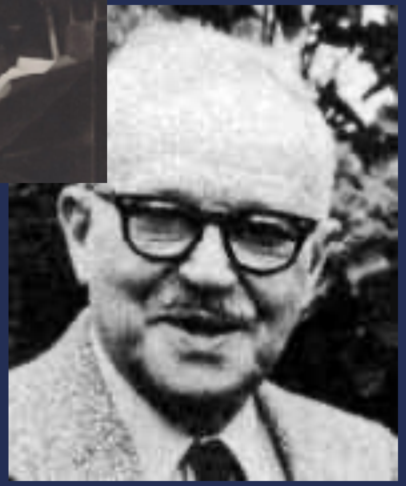
Karl Pearson

Ronald A Fisher







Jerzy Neyman

Egon Pearson

+ most of 20th century academic statisticians

Frequentist

Statistics Battle of the Century

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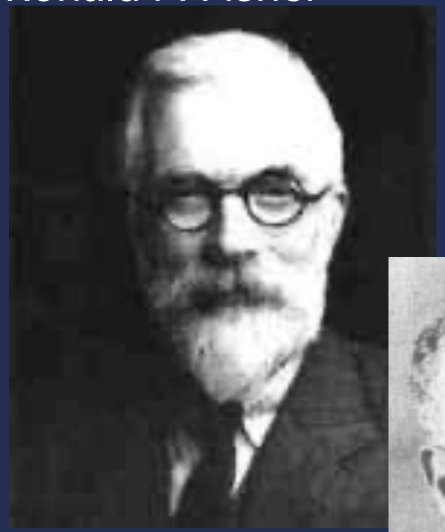

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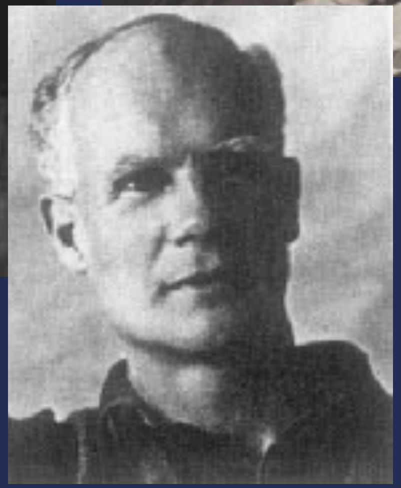

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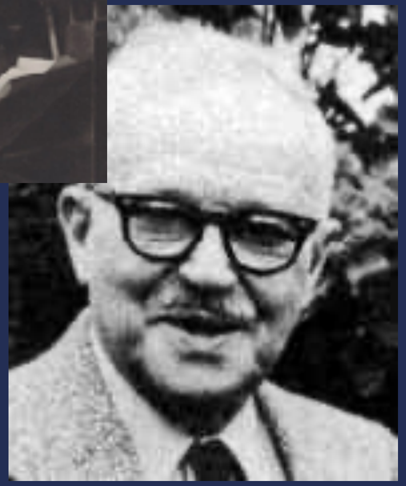

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vs

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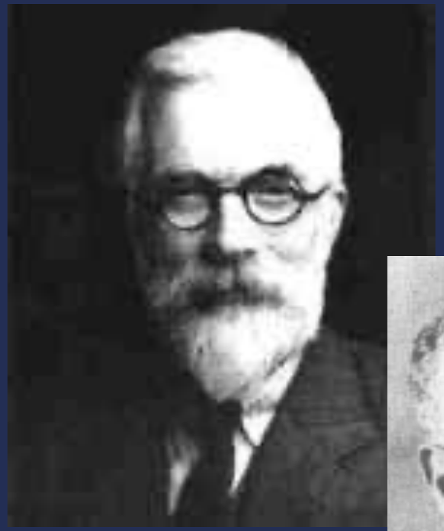

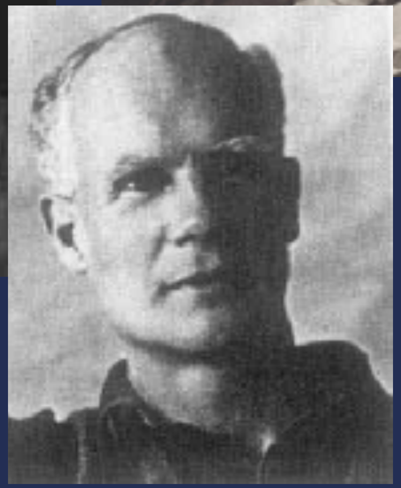

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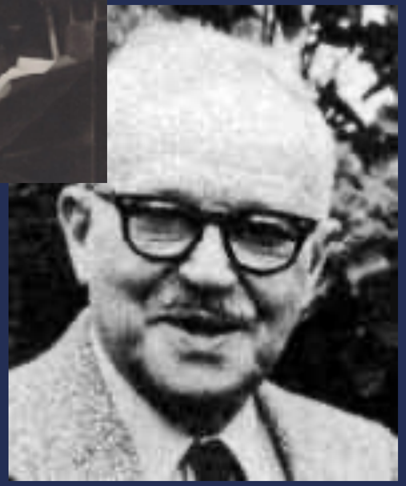

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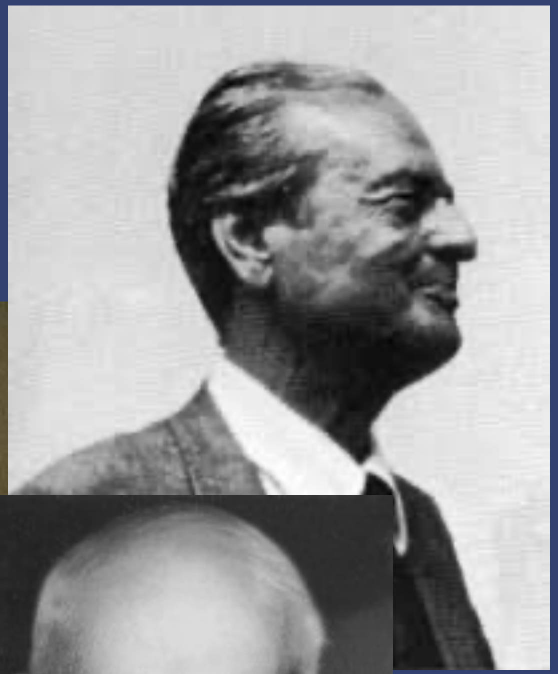
FREQUENTIST

VS

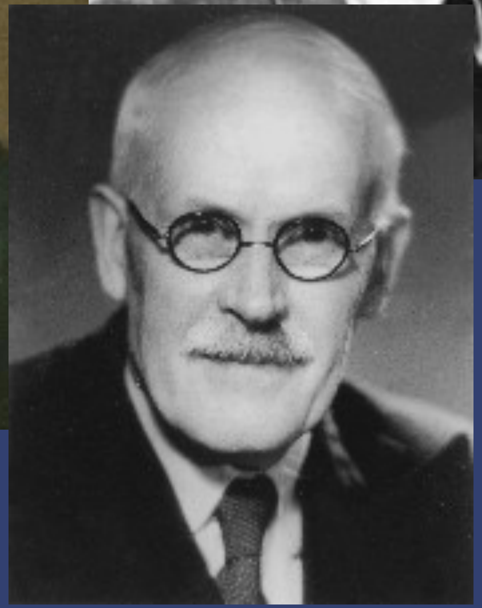
Thomas Bayes

Bruno de Finetti



Alan Turing

Pierre Simon
Maquis de Laplace



Harold Jeffreys

BAYESIAN

Frequentist

*except the mathematical assumptions of your statistical tests

Bayesian

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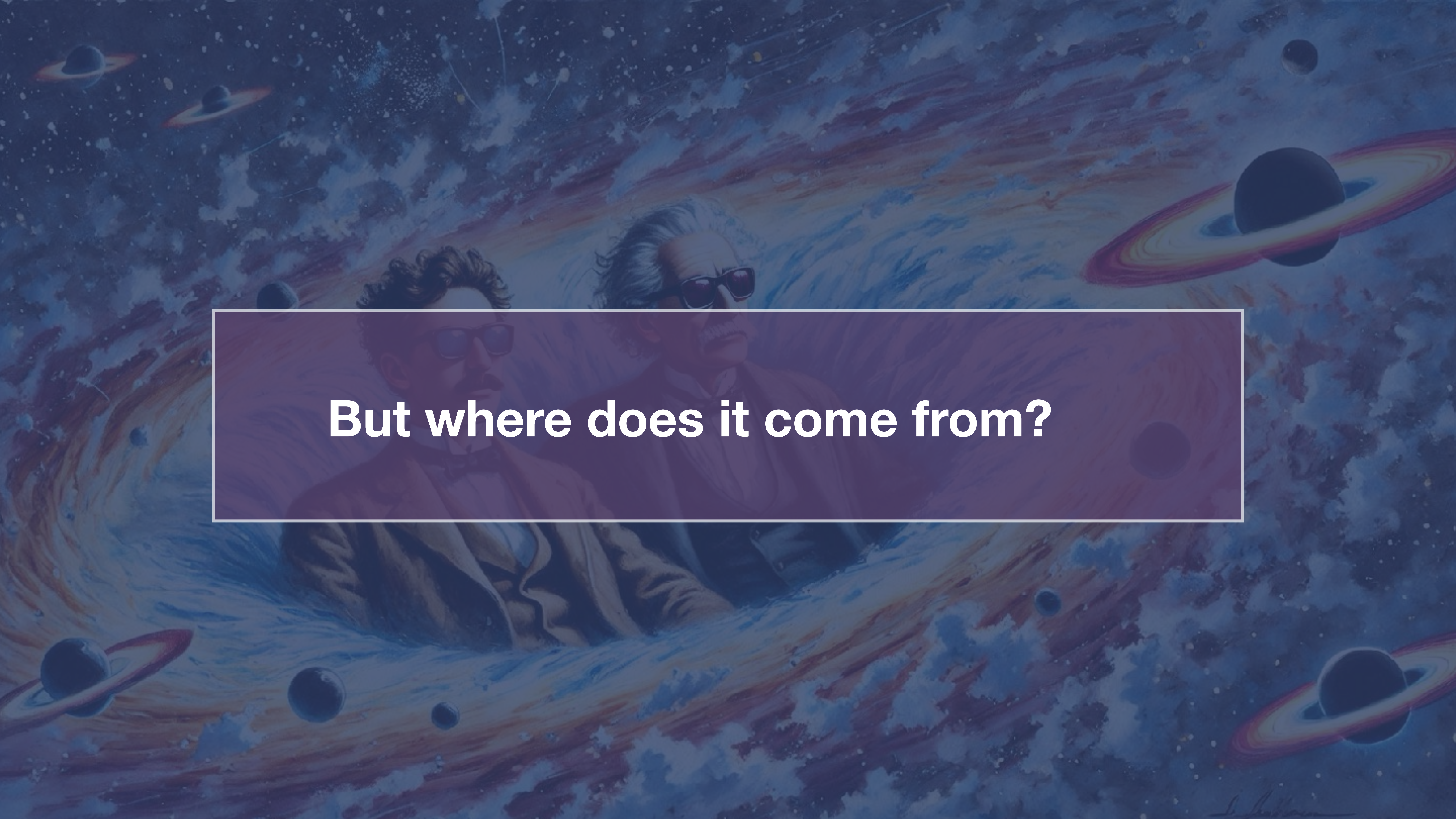
- Our belief in an outcome is a combination of our prior beliefs in that outcome, and the information supplied by the data.
- Enables direct computation of probabilities of hypotheses



But where does it come from?

(Conditional) Probabilities: Recap

Probability

Note: there exists more than one formalization of probability.

Most common: Kolmogorov formulation and Cox formulation

Probability is a (real-valued) set function that assigns to each event A in the sample space $\mathbf{S}$ a number $P(A)$, called the probability of the event A, such that the following hold:

1. The probability of any event must be nonnegative, that is, $P(A) \geq 0$.
2. The probability of the sample space is 1, that is, $P(\mathbf{S}) = 1$.
3. Given mutually exclusive events $A_1, A_2, A_3, \ldots$ that is, where $A_i \cap A_j = \emptyset$, for $i \neq j$,
   - the probability of a finite union of the events is the sum of the probabilities of the individual events, that is:
     $P(A_1 \cup A_2 \cup \ldots \cup A_k) = P(A_1) + P(A_2) + \ldots + P(A_k)$

More Probability

Probability of the Null:

If A is contained in B, then

Probability an event A is between 0 and 1

Probability of the complement of A is

Probability of A or B is

Notation

Probability of S

Probability of A

Probability of A and B

Probability of A or B

Probability of A given B

Probability of not A

Conditional Probabilities

- P(Astronomy) = ?
- P(2 | GRAPPA) = ?
- P(TP | 3) = ?
- Is P(1 | Astro) = P(Astro | 1) ?
- $\sum$P(Other | year) = ?

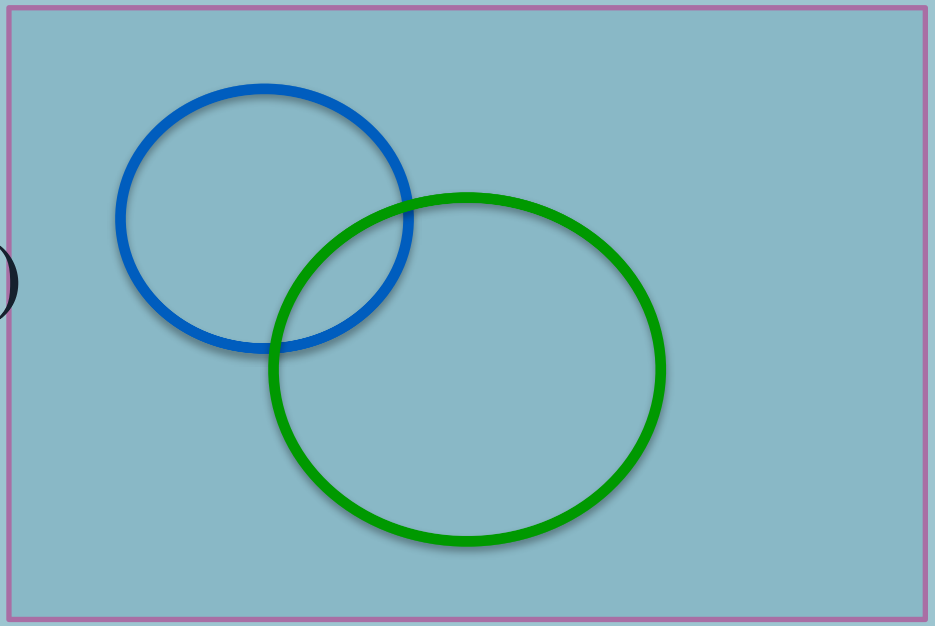
Suppose that this class has the following distribution of students:

Year/Track	Astronomy	GRAPPA	Theoretical Physics	Other	Total
1	19	9	2	3	33
2	3	2	1	0	6
3	1	0	1	2	4
Total	23	11	4	5	43

The probability of an event given some other event:

$$P(B\,|\,A) = \frac{P(A,B)}{P(A)}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Important: p(☔ | ☁️) ≠ p(☁️ | ☔)

Bayes Theorem

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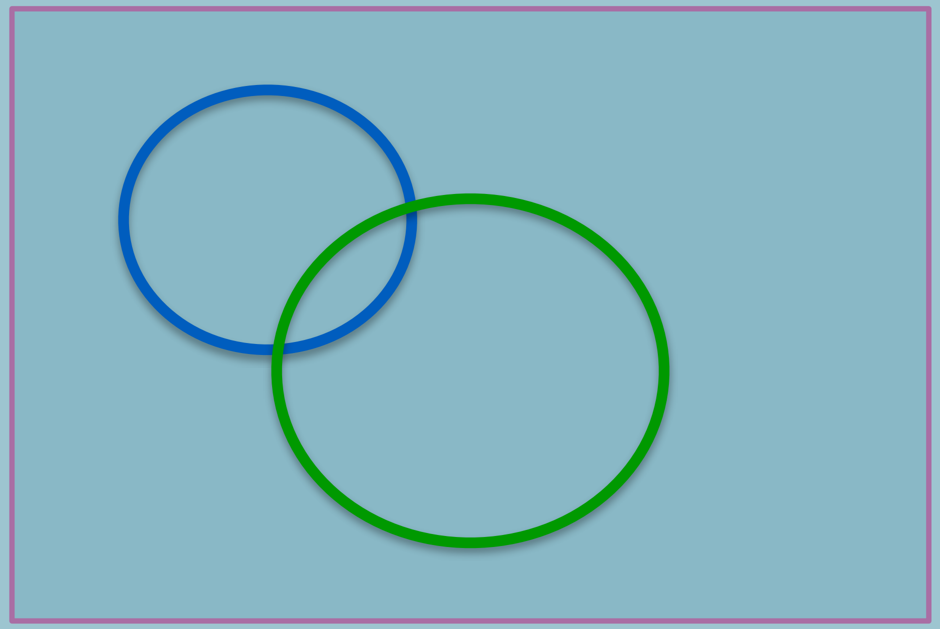
What is $P(A\,|\,B)$?



Questions

$$\int p(x \mid y)\,dx =$$

$$\int p(x \mid y)\,dy =$$

Questions

$$\int p(x \mid y)dx = \quad 1$$

$$\int p(x \mid y)dy =$$

Questions

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$$\int p(x \mid y)\,dy = \quad$$ 👎

Multiplication Rule

$$p(x \mid y) = p(x, y)/p(y)$$

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$$p(x) = \int_{-\infty}^{\infty} p(x, y)dy = \int_{-\infty}^{\infty} p(x \mid y)p(y)dy$$

$$p(x \mid y) = p(x, y)/p(y)$$

$$p(x, y) = p(x \mid y)p(y)$$

$$p(y \mid x) = ?$$

Marginalisation

$$p(x \mid y) = p(x, y)/p(y)$$

$$p(x, y) = p(x \mid y)p(y)$$

$$p(y \mid x) = ?$$

$$p(x, y) = p(x \mid y) p(y)$$

$$p(x, y) = p(y \mid x) p(x)$$

Bayes' Theorem

$$p(x, y) = p(x \mid y)p(y)$$

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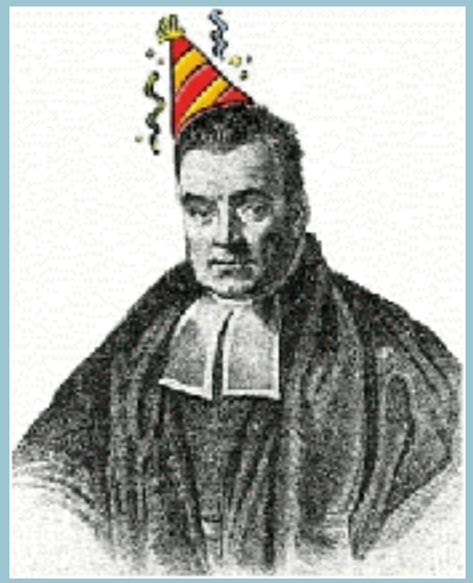
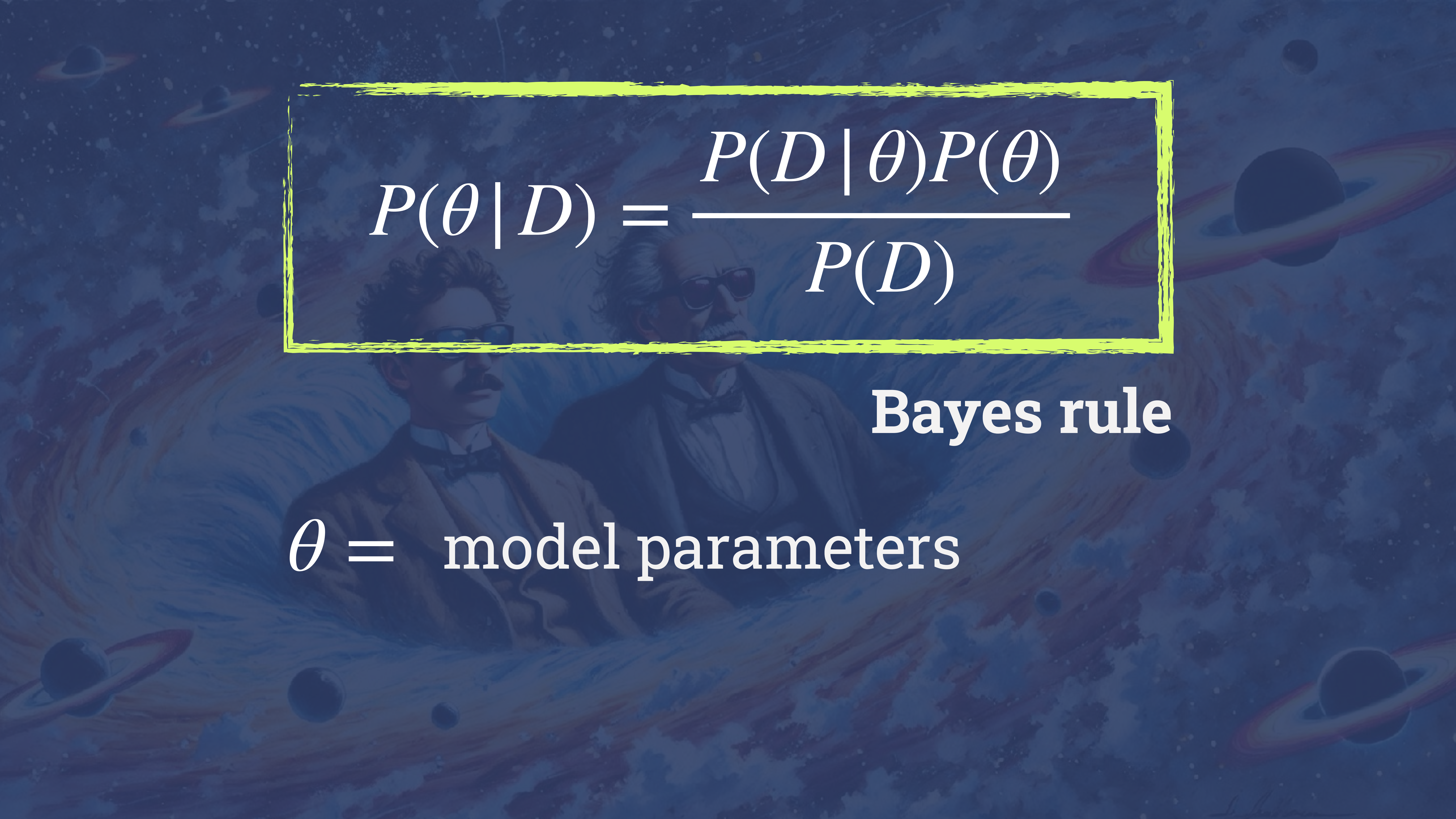
$$\boxed{p(x \mid y)p(y) = p(y \mid x)p(x)}$$



$$\boxed{P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}}$$

Bayes rule



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Bayes rule

$\theta =$ model parameters

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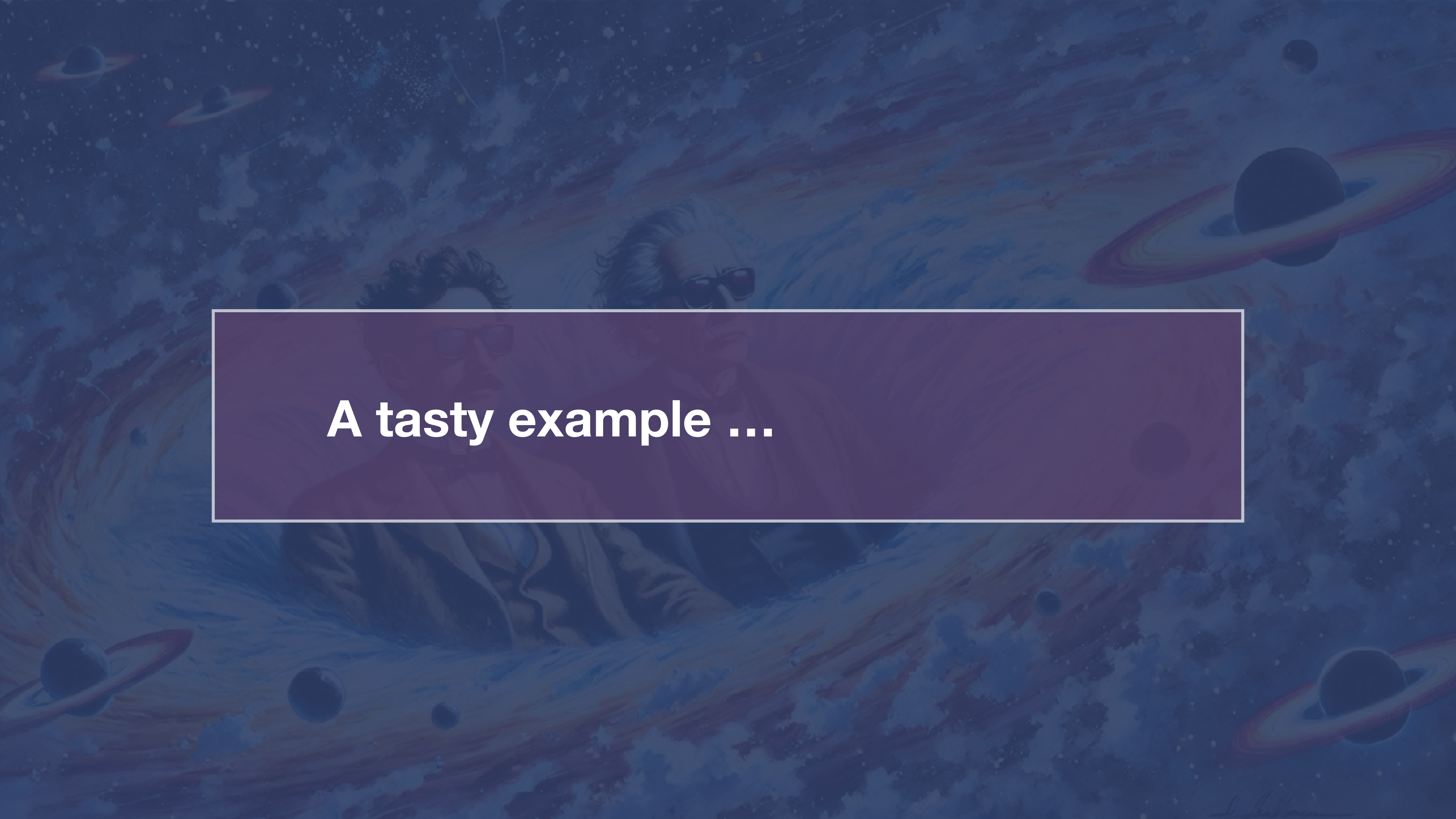
Bayes rule

$\theta =$ model parameters

$D =$ data


A tasty example …

posterior

likelihood

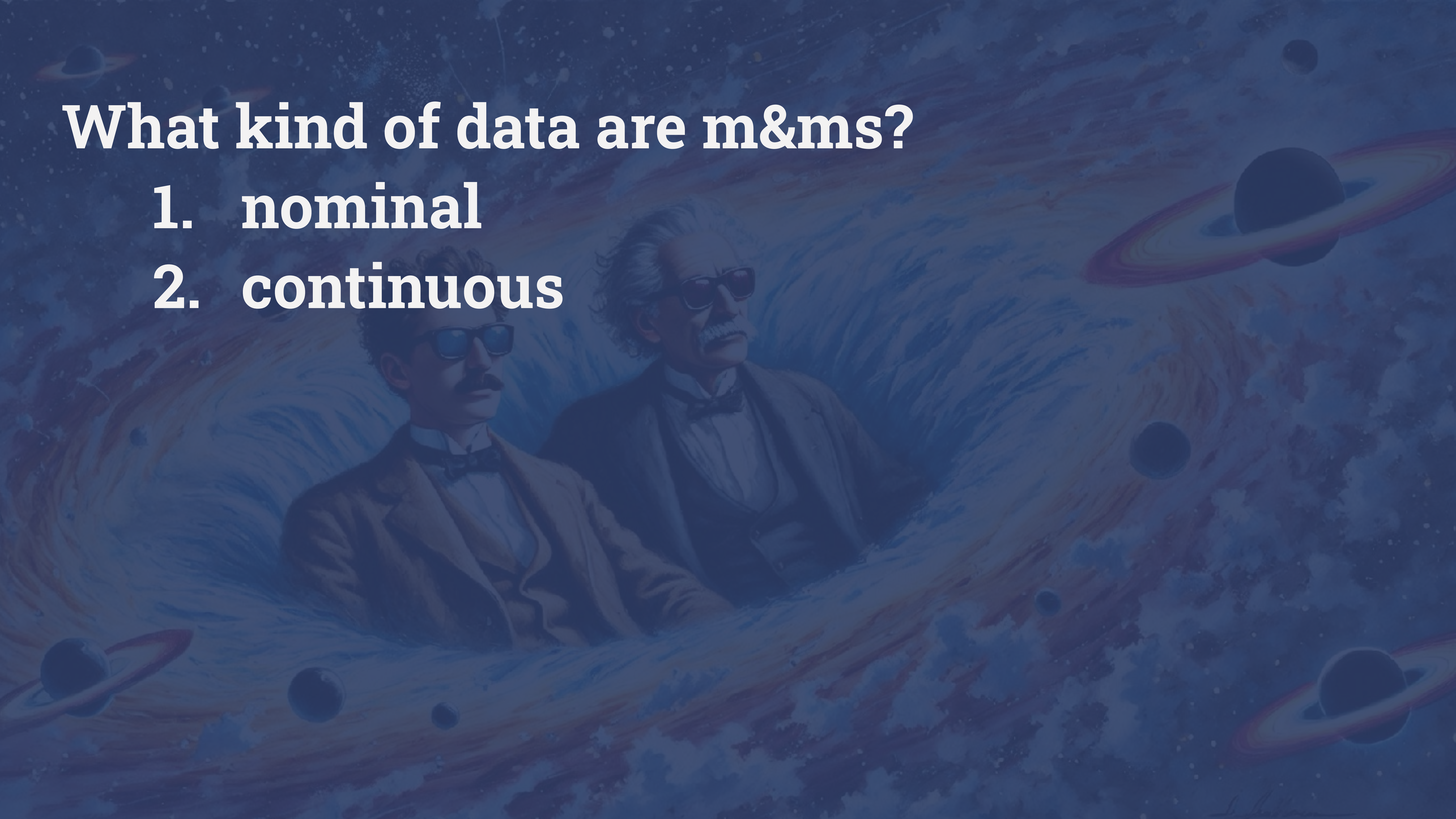
prior

$$P(\theta \mid D) \propto P(D \mid \theta) P(\theta)$$

Find the posterior distribution for the percentage of blue m&ms made at the factory


PEANUT
CHOCOLATE CANDIES
m&m's
250
CALORIES
NET WT 1.74 OZ (49.3g)

Question: Why do **you** think there are sometimes more or fewer blue M&Ms in the package?



What kind of data are m&ms?

1. nominal
2. continuous

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How will you record the data?

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How will you record the data?

How might we model the probability of drawing a blue m&m?

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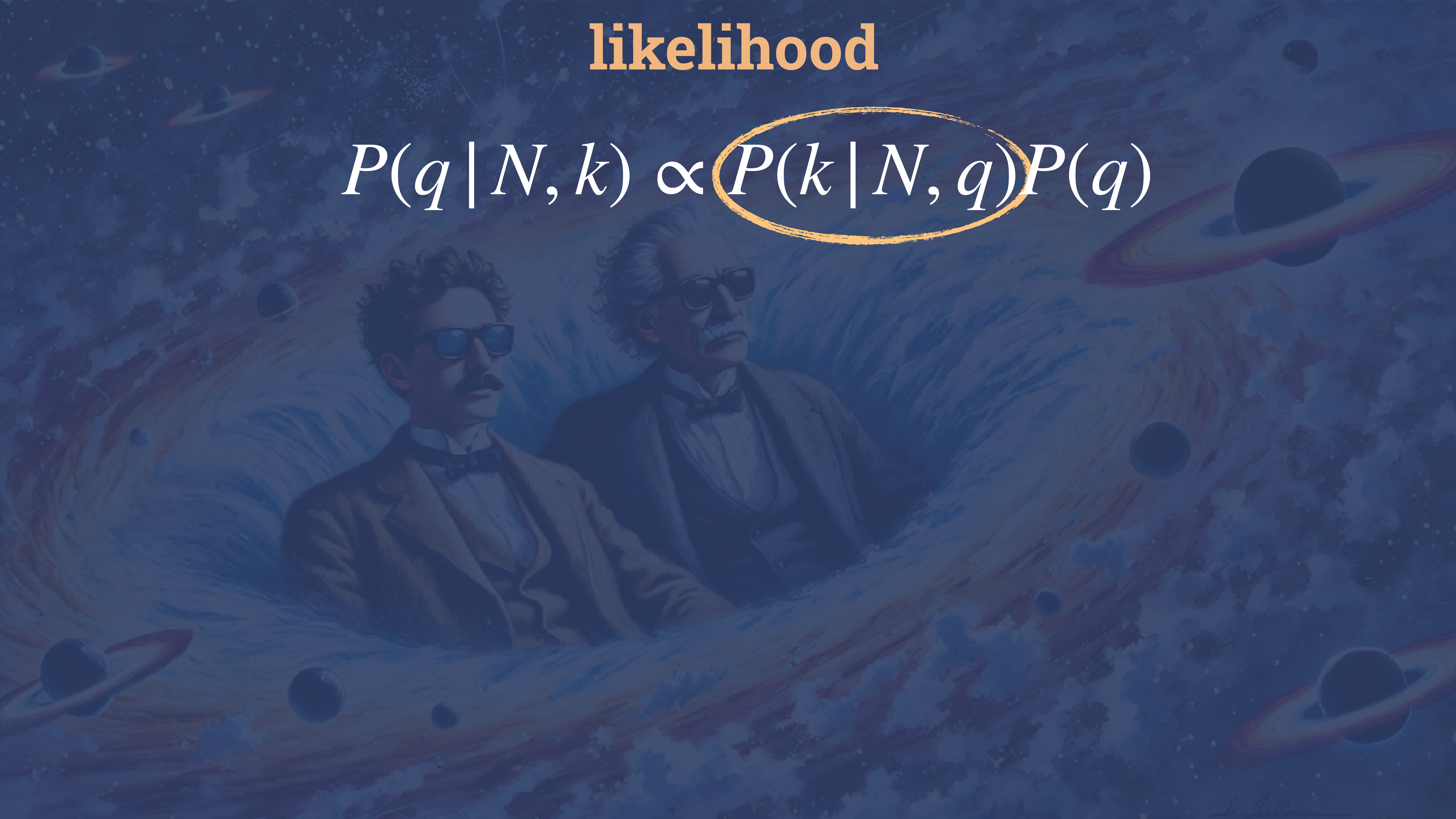
How will you record the data? Table (all colours)

How might we model the probability of drawing a blue m&m?

likelihood

$$P(q \mid N, k) \propto P(k \mid N, q) P(q)$$



likelihood

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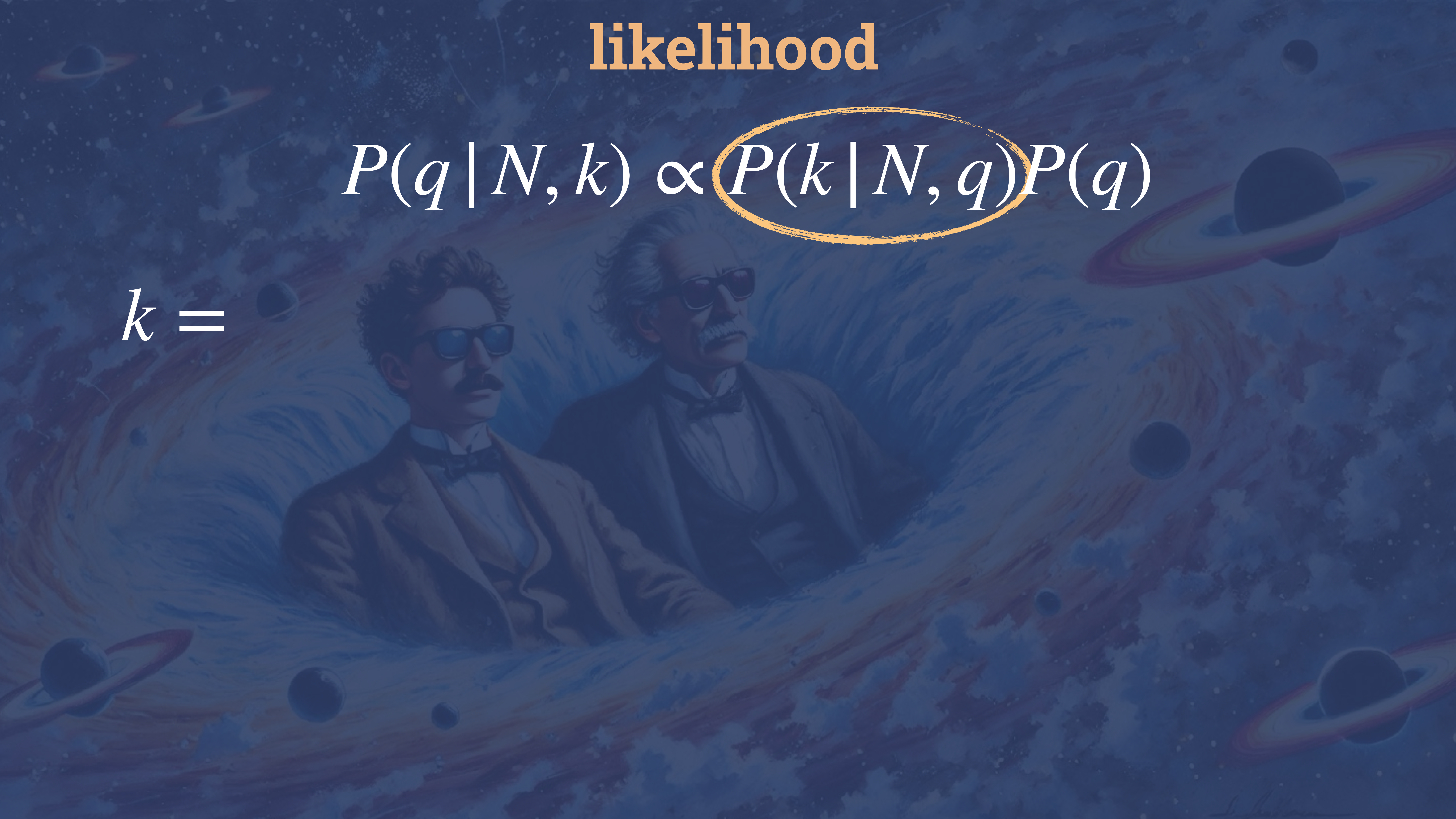
$$k =$$



likelihood

$$P(q \mid N, k) \propto P(k \mid N, q) P(q)$$

$k =$ number of successes (blue m&ms)

likelihood

$$P(q \mid N, k) \propto P(k \mid N, q) P(q)$$

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$N - k =$

likelihood

$$P(q\,|\,N, k) \propto P(k\,|\,N, q)P(q)$$

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likelihood

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$q =$ probability of drawing a blue m&m

likelihood

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$k =$ number of successes (blue m&ms)

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$q =$ probability of drawing a blue m&m

$P(\theta \mid D) \propto P(D \mid \theta) P(\theta)$ with $D = k$, $\theta = q$

The Binomial Distribution

Observed fraction of blue M&Ms in a single bag

True fraction of blue M&Ms produced at the factory

$$p(D\,|\,\theta) = p(\,k\,|\,N,q\,) \sim \mathrm{Binom}(N, k, q)$$

"The probability of observing k blue M&Ms out of N given that the true fraction of M&Ms is q"

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$$= \binom{N}{k} q^k (1-q)^{(N-k)}$$

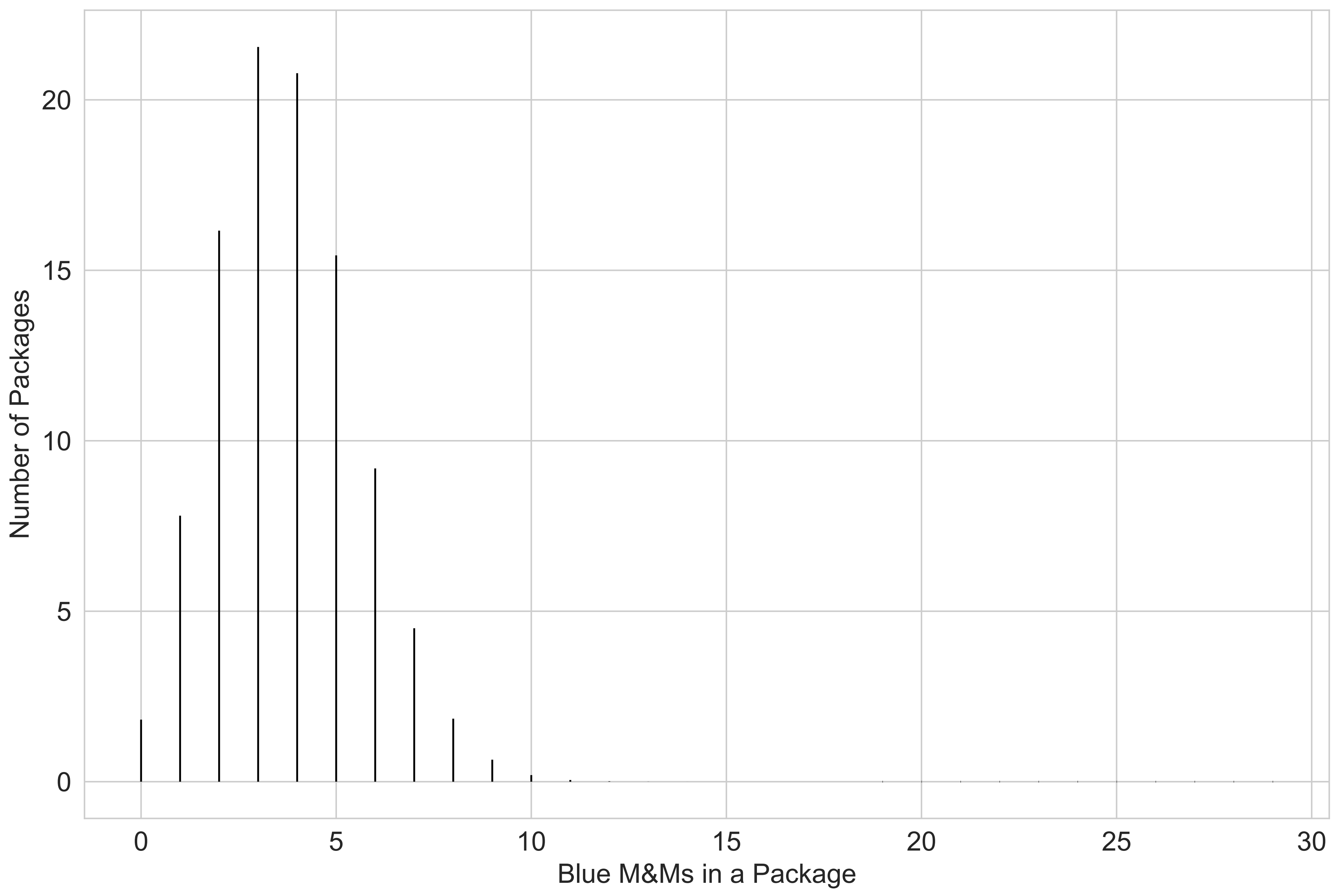
"The probability of observing k blue M&Ms out of N given that the true fraction of M&Ms is q"

PEANUT
m&m's
FUN SIZE

What do you think the fraction of blue M&Ms produced at the factor should be?

First Assumption: The M&Ms are mixed together at the factory and then distributed into packages.

Second Assumption: All colours are equally distributed



Binomial distribution

Blue Fraction: 0.125

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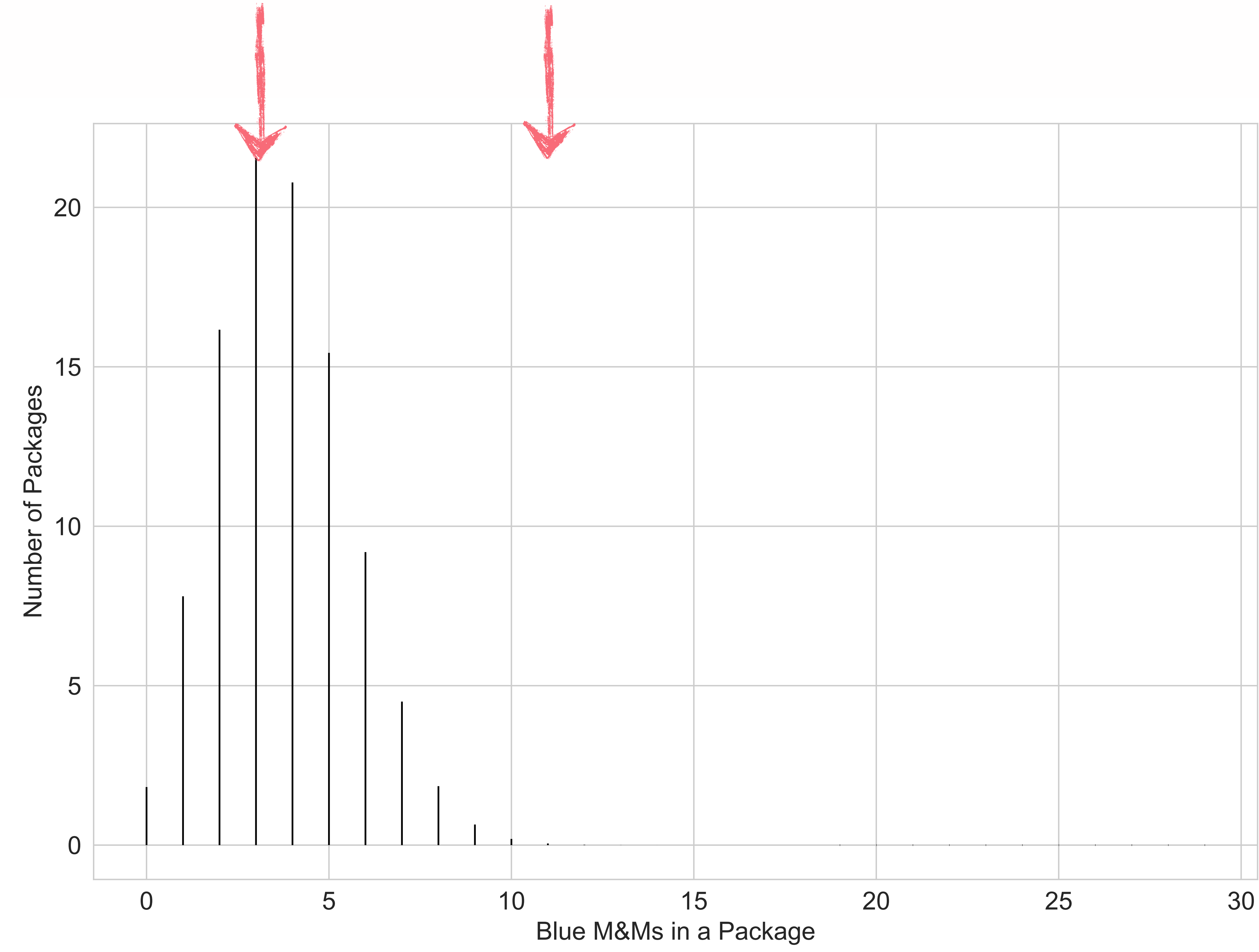

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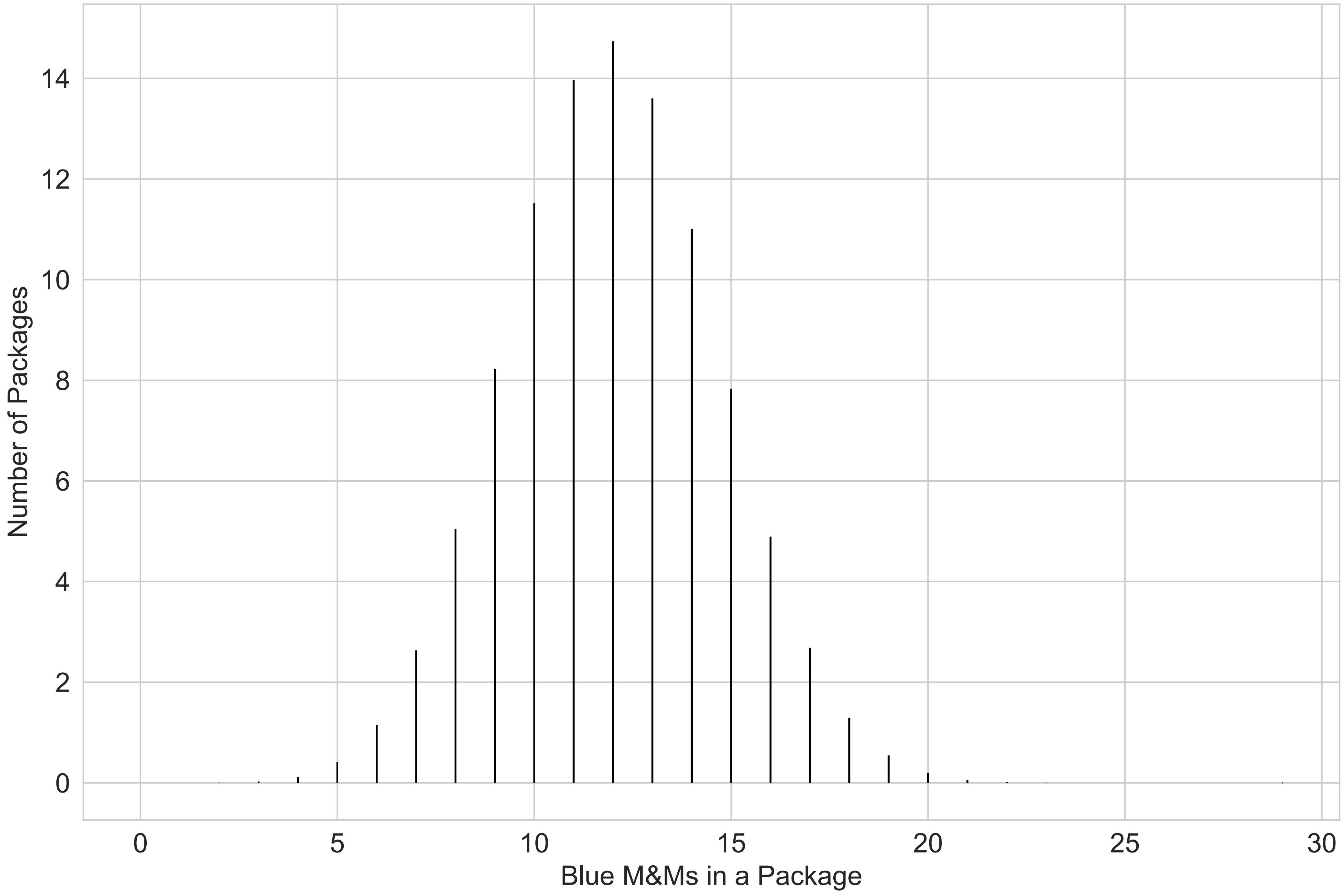

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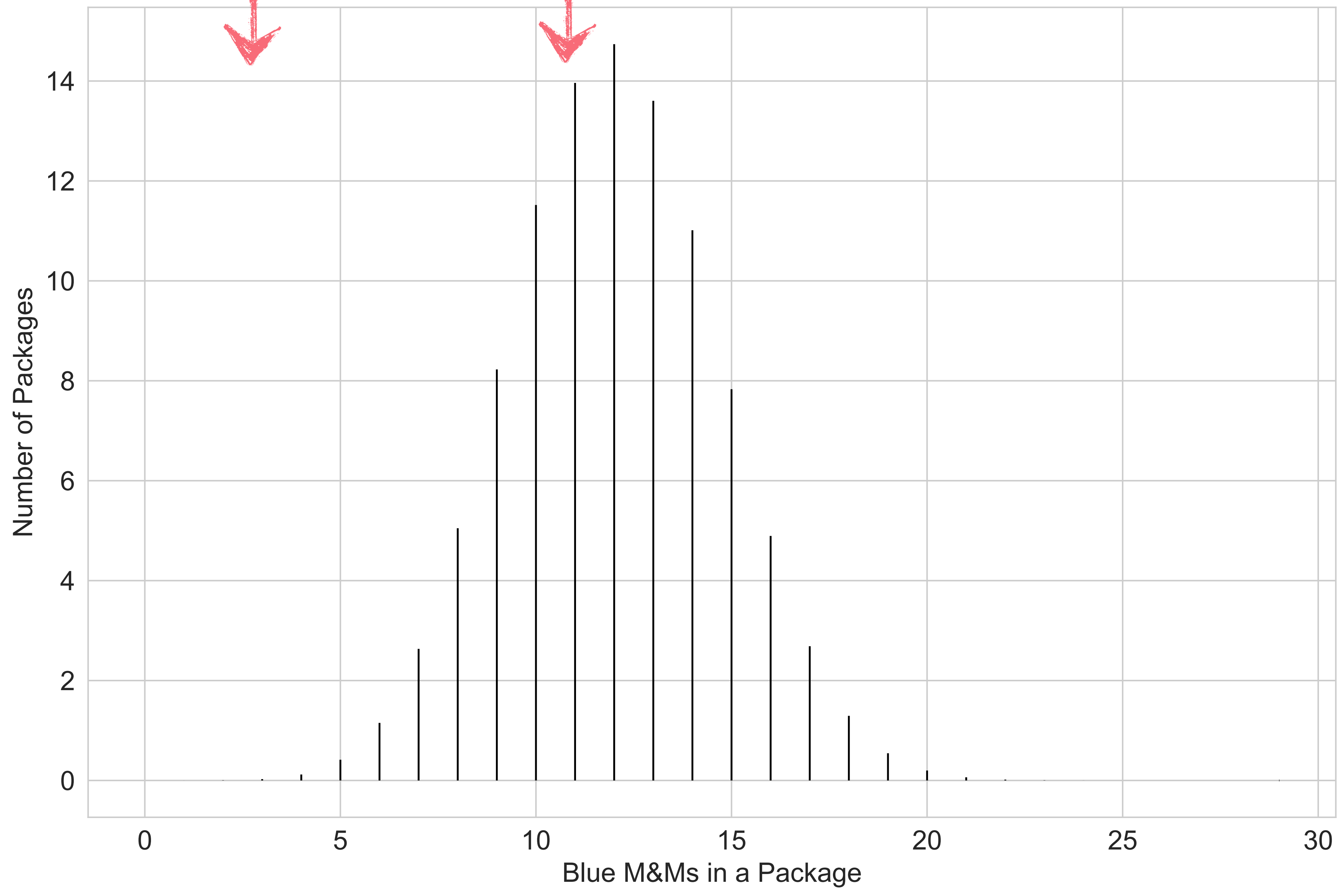


Binomial distribution

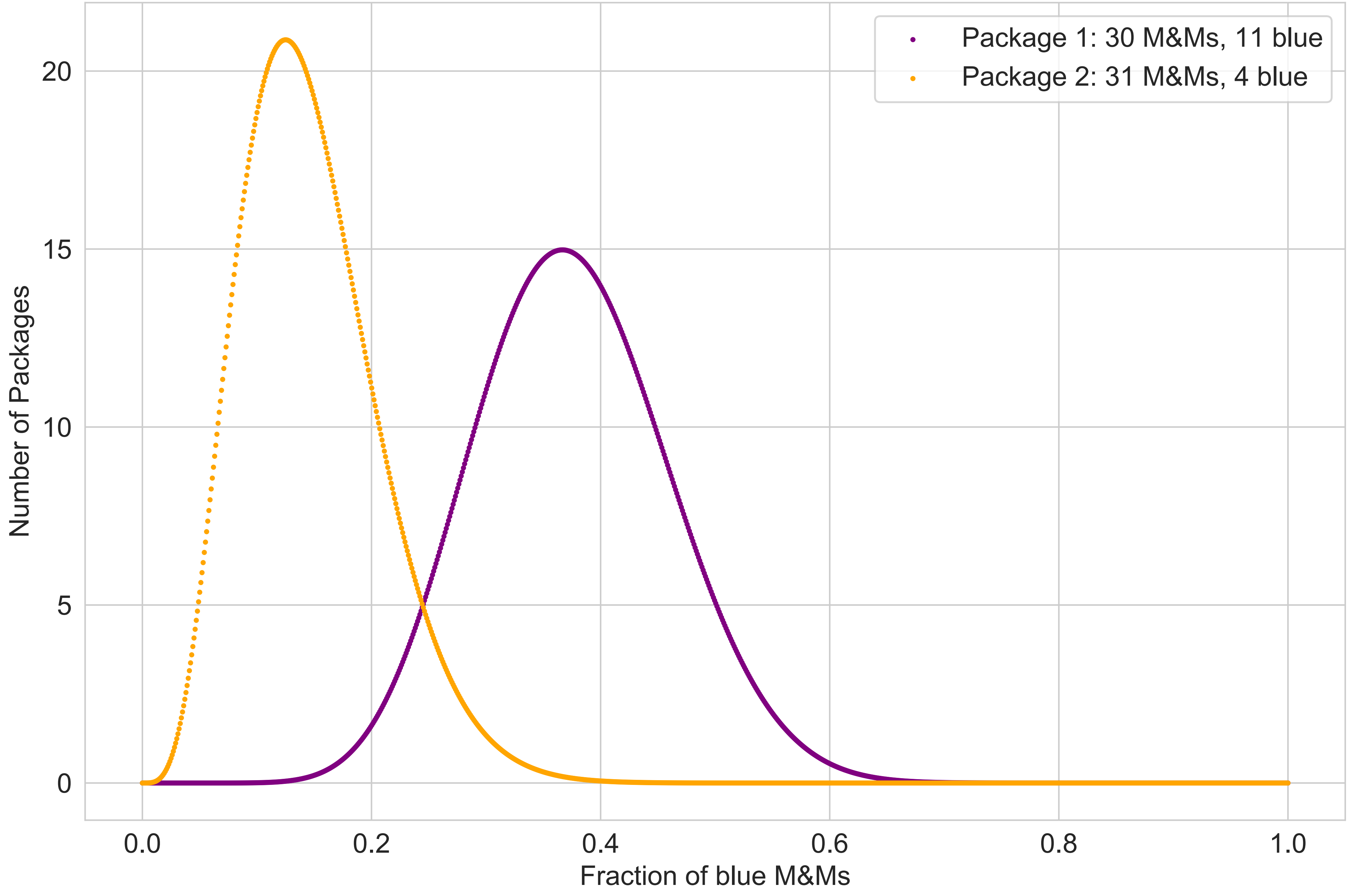
Blue Fraction: 0.4

Question: Can we use this somehow to estimate the fraction of blue M&Ms?

https://github.com/philuttley/stats-methods-25/blob/gh-pages/_extras/Modelling_M&Ms.ipynb

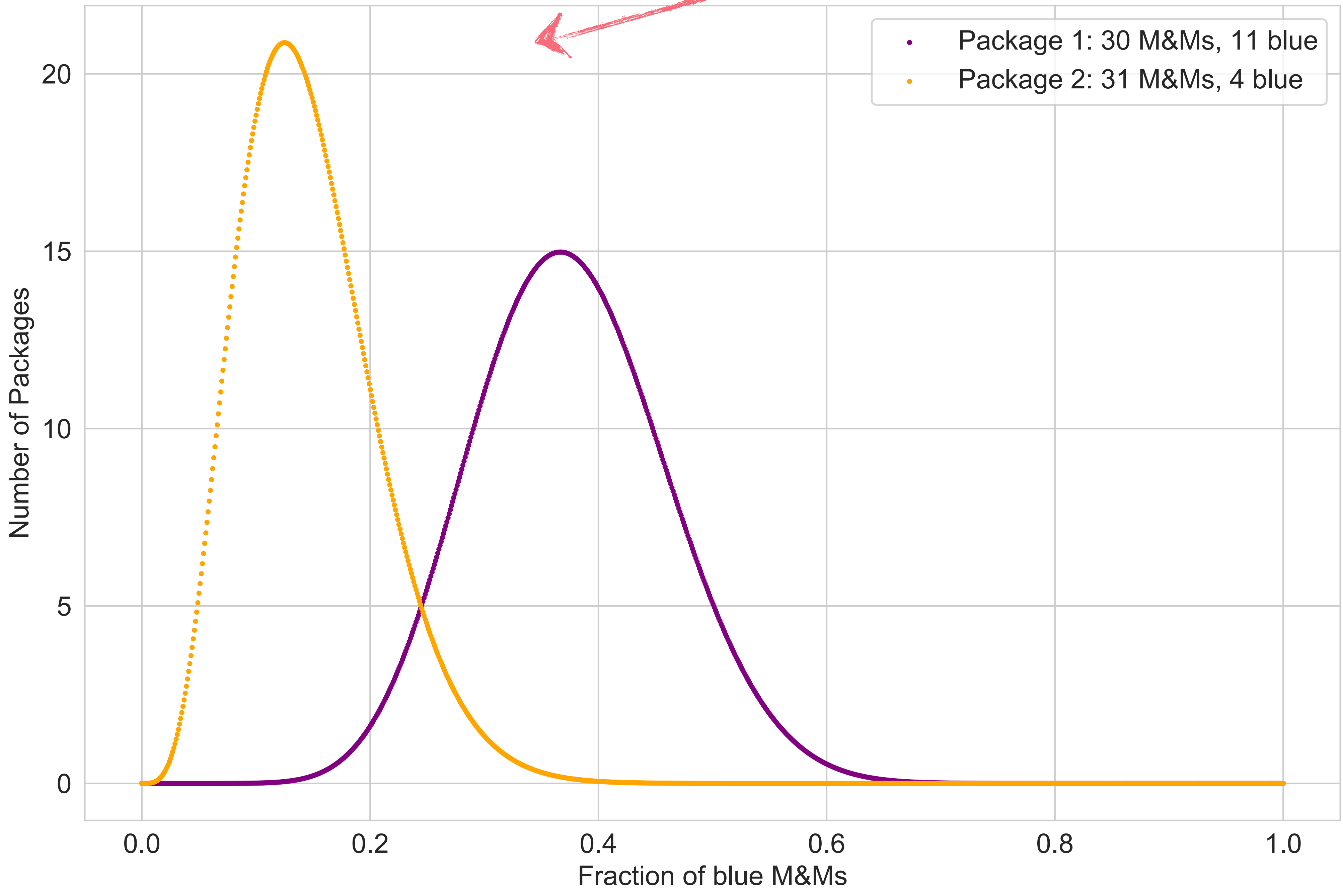
Exercises 1, 2, 3

The Likelihood Function



The Likelihood Function

use optimization routines to find the maximum



Problem: The Likelihood assumes we know the **true underlying fraction** of M&Ms.

We're interested in $p(q \mid k, N)$, not $p(k \mid q, N)$

Conditional Probability Magic

likelihood prior

posterior

$$p(q \mid k, N) = \frac{p(k \mid q, N)\, p(q)}{p(k \mid N)}$$

Evidence/marginal likelihood

Bayes theorem

Conditional Probability Magic

$$\underbrace{p(q \mid k, N)}_{\text{posterior}} = \frac{\overbrace{p(k \mid q, N)}^{\text{likelihood}}\ \overbrace{p(q)}^{\text{prior}}}{\underbrace{p(k \mid N)}_{\text{Evidence/marginal likelihood}}}$$

$$p(q \mid k, N) \propto p(k \mid q, N)\, p(q)$$

Bayes theorem

Question

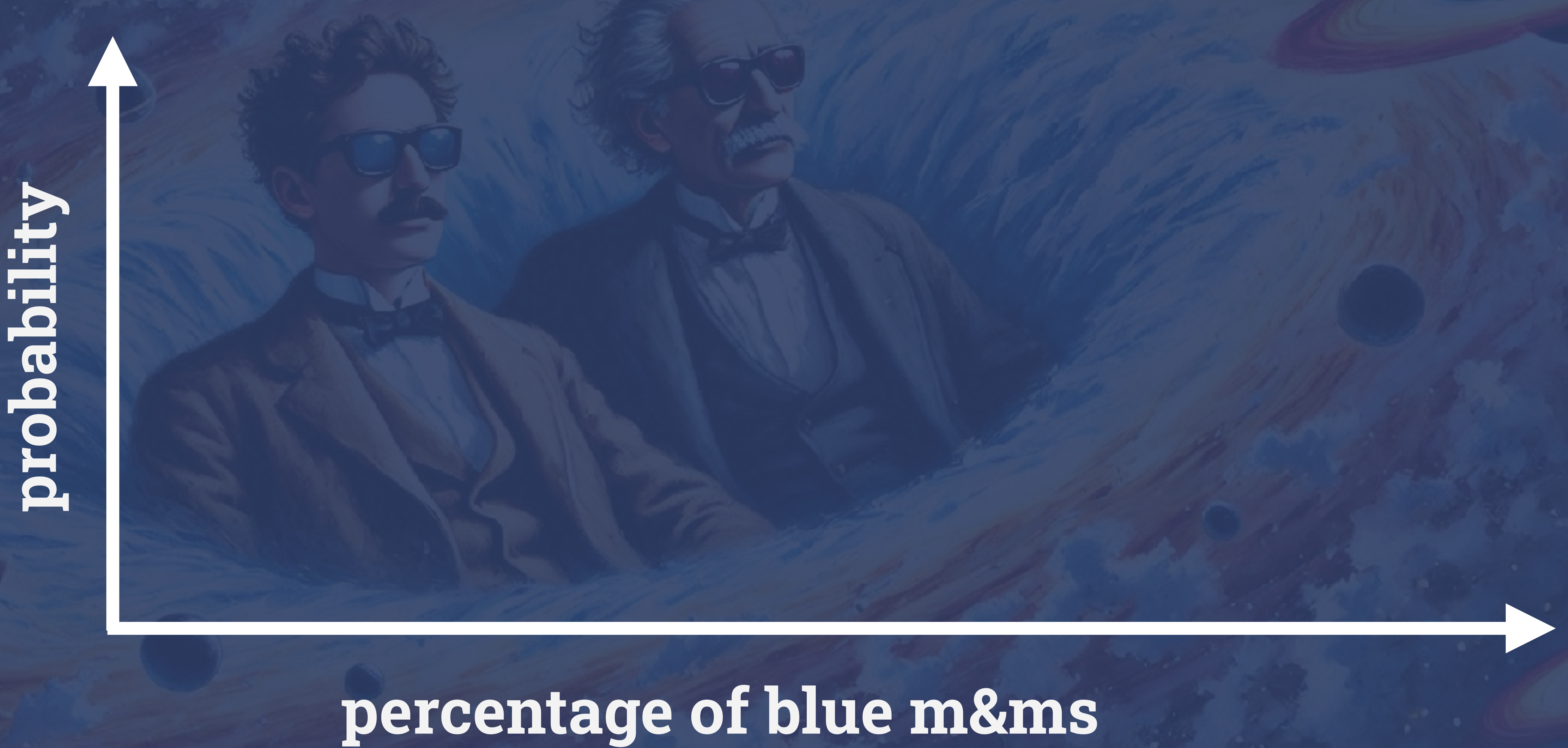
What prior information do we have about $p(f)$?

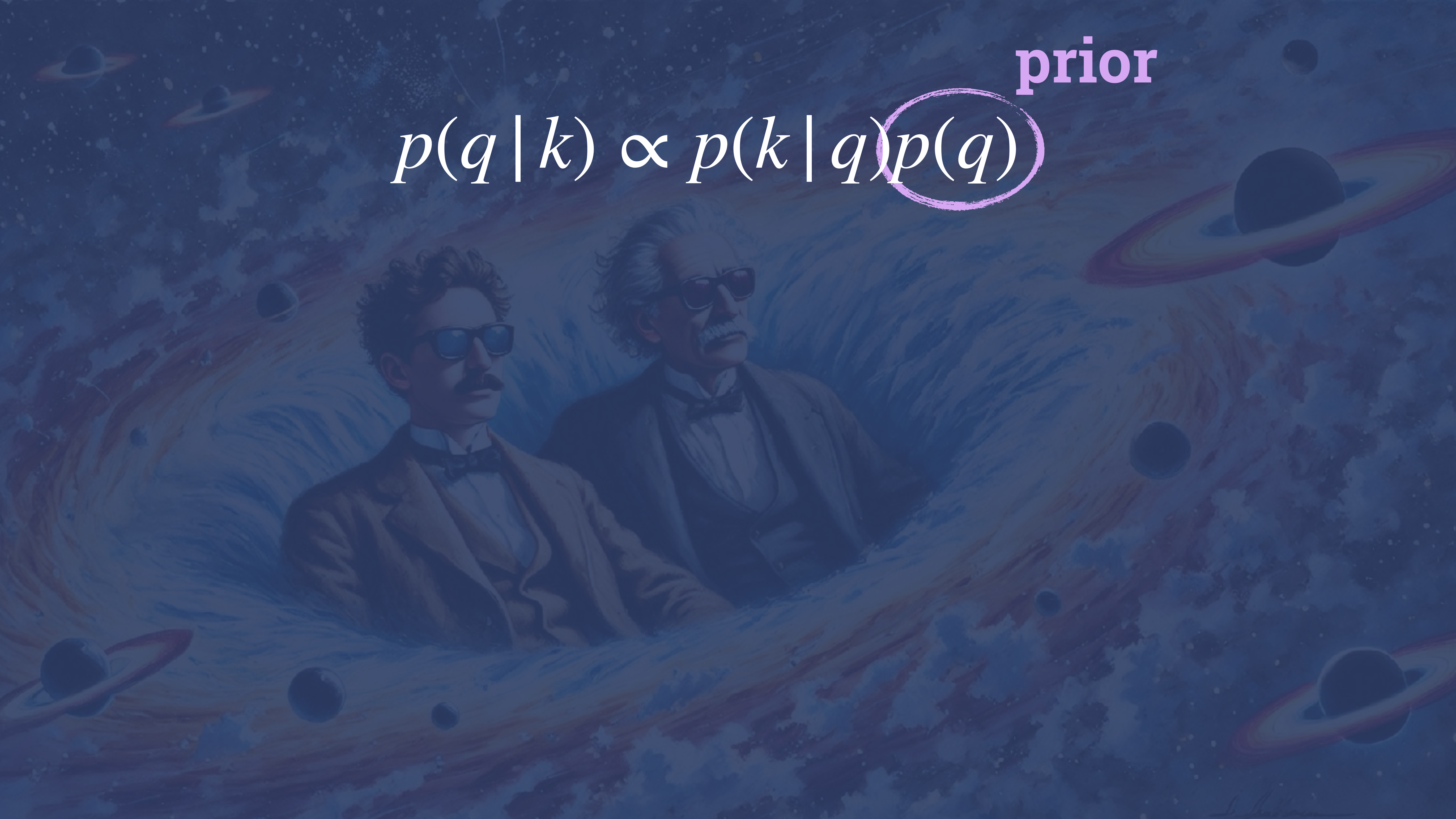
How many different colours of M&Ms are there?

What percentage of blue m&ms are made at the factory?

Do you think every bag of m&ms will have the same percentage of blue m&ms?

Sketch your prior




prior
$p(q\,|\,k) \propto p(k\,|\,q)p(q)$

prior

$$p(q \mid k) \propto p(k \mid q)\, p(q)$$

$$\boxed{p(q) \propto q^{\alpha-1}(1-q)^{\beta-1}}$$

$$p(q \mid k) \propto p(k \mid q) p(q)$$

prior

beta distribution

$$\boxed{p(q) \propto q^{\alpha-1}(1-q)^{\beta-1}}$$

prior

$$p(q \mid k) \propto p(k \mid q) p(q)$$

beta distribution

$$\boxed{p(q) \propto q^{\alpha-1}(1-q)^{\beta-1}}$$

“conjugate prior”

$$p(q \mid k) \propto p(k \mid q) p(q)$$

prior

beta distribution

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“conjugate prior”

$$\alpha = ?$$

prior

$$p(q\,|\,k) \propto p(k\,|\,q)p(q)$$

beta distribution

$$\boxed{p(q) \propto q^{\alpha-1}(1-q)^{\beta-1}}$$

“conjugate prior”

$$\alpha = ? \qquad \beta = ?$$

Write a function for the beta distribution and try out different values for alpha and beta

Some values to try:

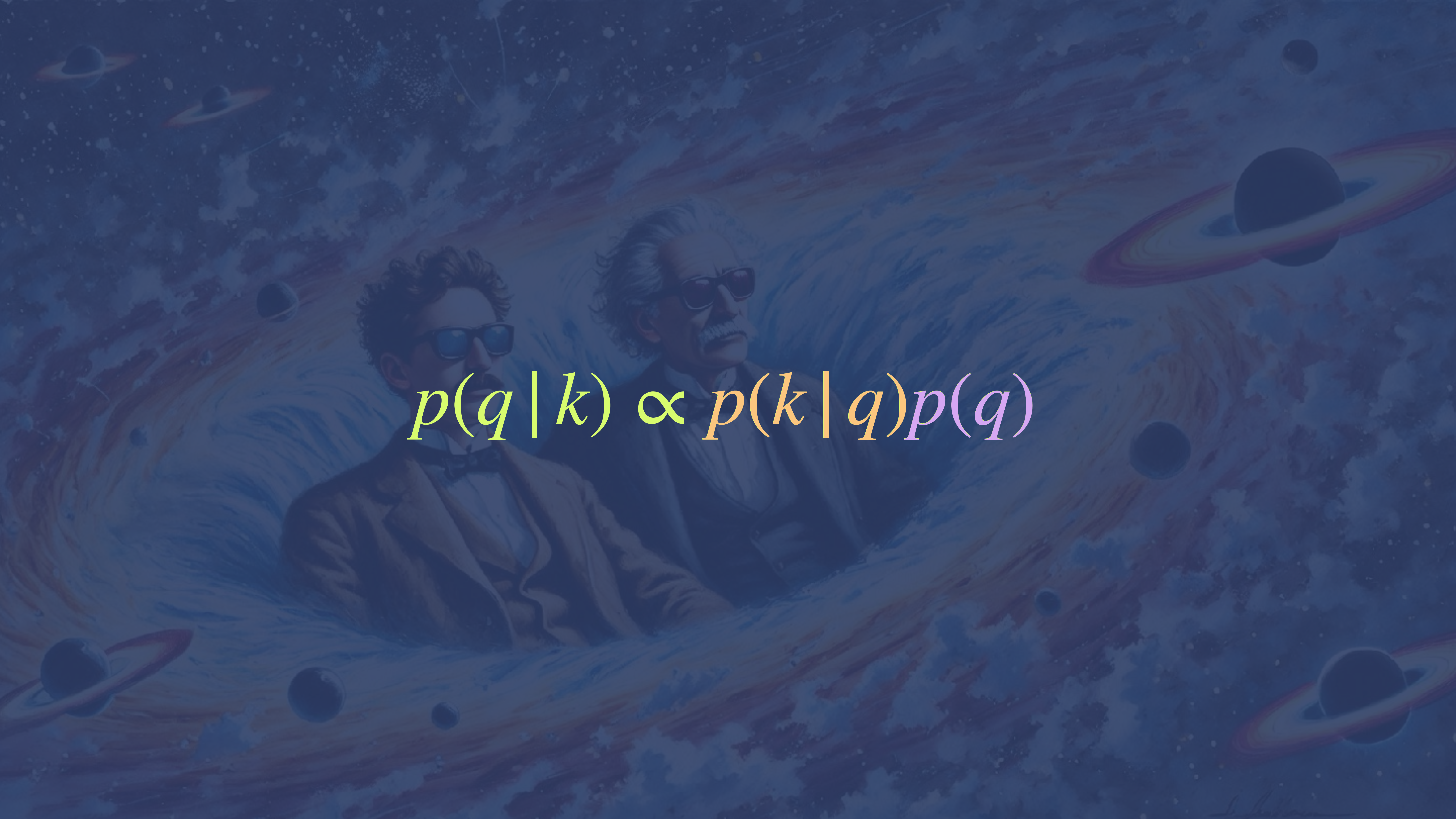
$$\alpha = \beta$$

$$\alpha = \beta = 1$$

$$\alpha, \beta \quad \text{very large}$$

$$\alpha, \beta \quad \text{very small}$$

https://github.com/philuttley/stats-methods-25/blob/gh-pages/_extras/Modelling_M&Ms.ipynb Exercise 5 + 6



$$p(q \mid k) \propto p(k \mid q)p(q)$$

likelihood

$$\boxed{p(k \mid q, N) \propto q^k (1-q)^{N-k}}$$

$$p(q \mid k) \propto p(k \mid q)\, p(q)$$

likelihood

$$\boxed{p(k\,|\,q, N) \propto q^k(1-q)^{N-k}}$$

prior

$$\boxed{p(q) \propto q^{\alpha-1}(1-q)^{\beta-1}}$$

$$p(q\,|\,k) \propto p(k\,|\,q)p(q)$$

likelihood

$$\boxed{p(k\,|\,q, N) \propto q^{k}(1-q)^{N-k}}$$

prior

$$\boxed{p(q) \propto q^{\alpha-1}(1-q)^{\beta-1}}$$

$$p(q\,|\,k) \propto p(k\,|\,q)\,p(q)$$

$$\boxed{p(q\,|\,k, N) \propto q^{k+\alpha-1}(1-q)^{N-k+\beta-1}}$$

Gather some data!

record the counts for all colours and the total number of M&Ms:

https://forms.gle/Dvm1jpbUdTT6M1Vs8

Use the function for the beta distribution to plot both the prior and the posterior in the same plot

- Is the posterior distribution what you expected?
- Compare the posterior distribution to the prior distribution: Is this the result you expected, given six different colours?
- How sensitive is the posterior to the prior?

Use the function for the beta distribution to plot both the **prior** and the **posterior** in the **same plot**

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https://github.com/philuttley/stats-methods-25/blob/gh-pages/_extras/Modelling_M&Ms.ipynb

Use the function for the beta distribution to plot both the **prior** and the **posterior** in the **same plot**

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https://github.com/philuttley/stats-methods-25/blob/gh-pages/_extras/Modelling_M&Ms.ipynb

Exercise 7+8

How would you expect the posterior to change with **more data**?

Let's **pool all the data** and find out!

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Let's **pool all the data** and find out!

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How would you expect the posterior to change with **more data**?

Let's **pool all the data** and find out!

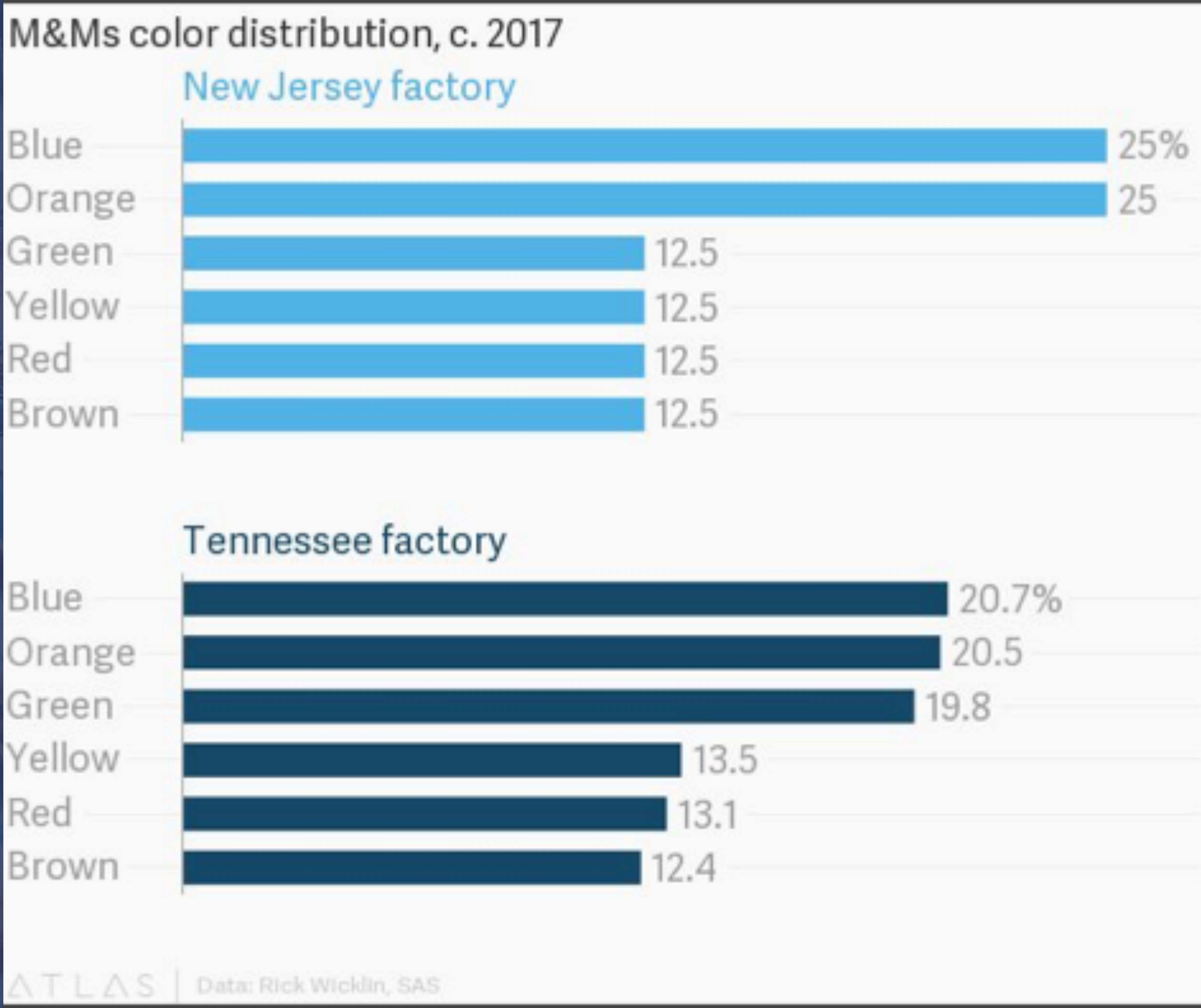
https://github.com/philuttley/stats-methods-25/blob/gh-pages/_extras/Modelling_M&Ms.ipynb

Exercise 9

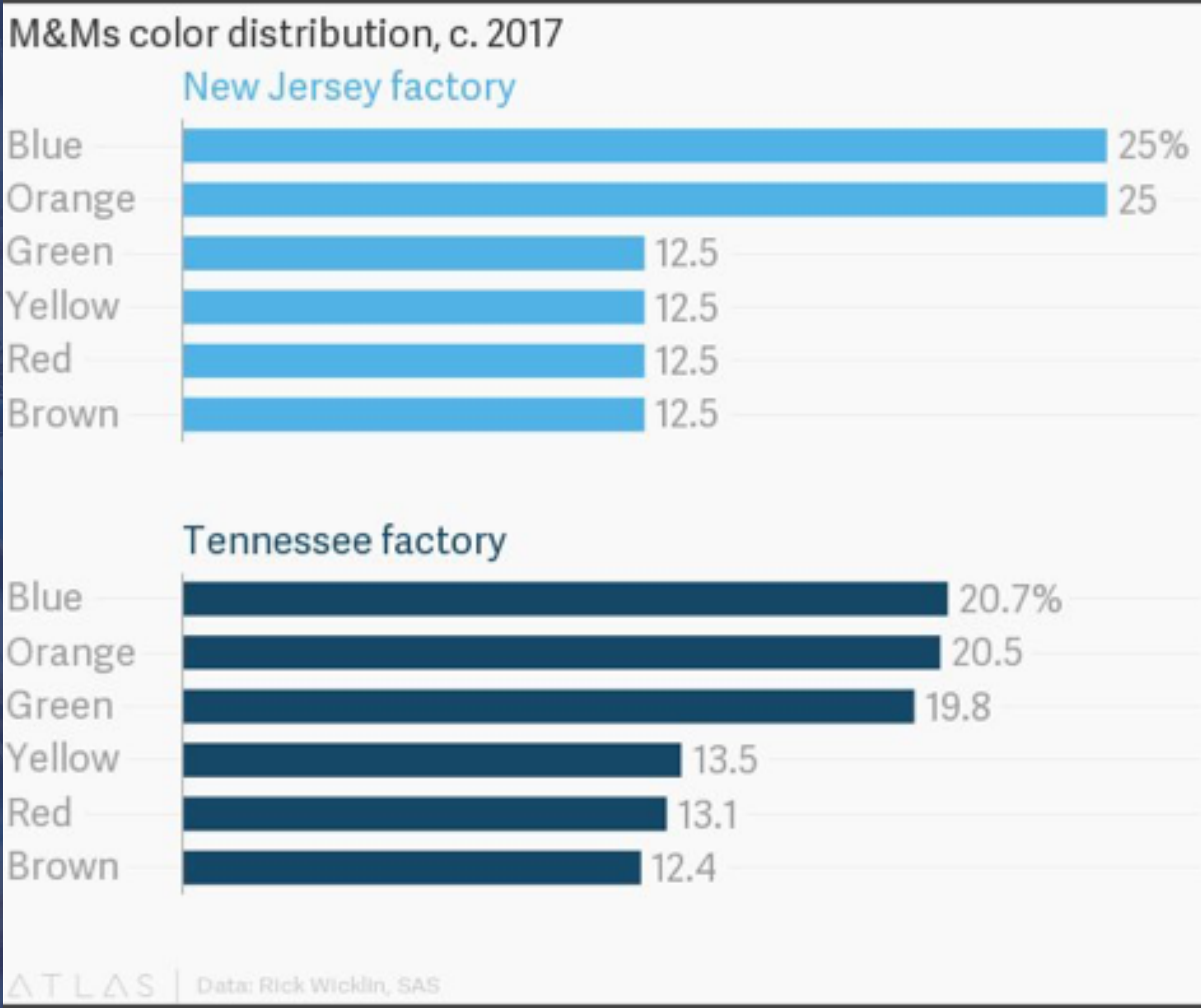
Priors are useful when data isn't very informative.

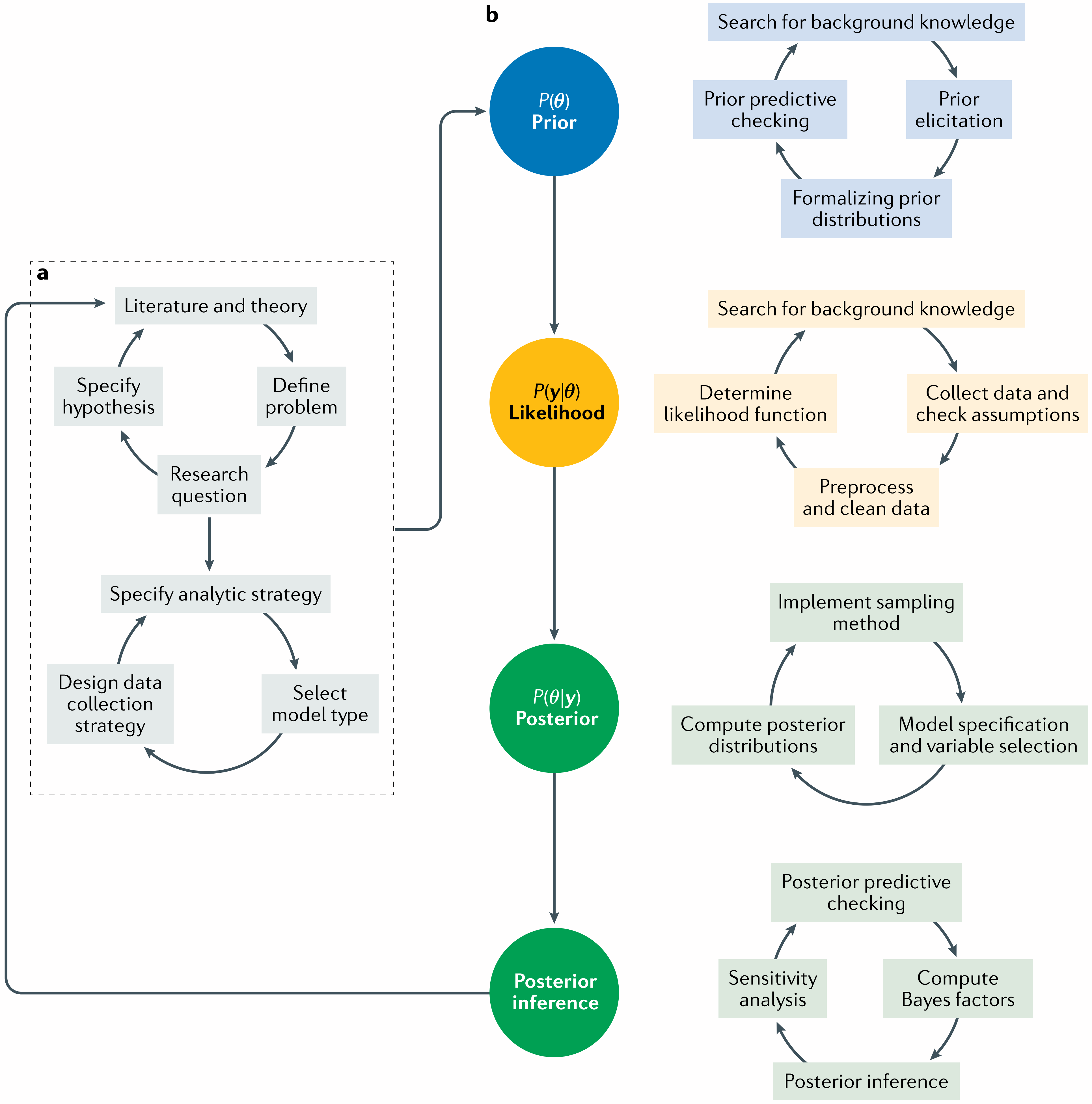
When the data is informative, the choice of prior doesn't matter much.

We kind of know the distributions of **US factories**, and different factories make different distributions of m&ms!



Do **European factories** produce similar proportions?



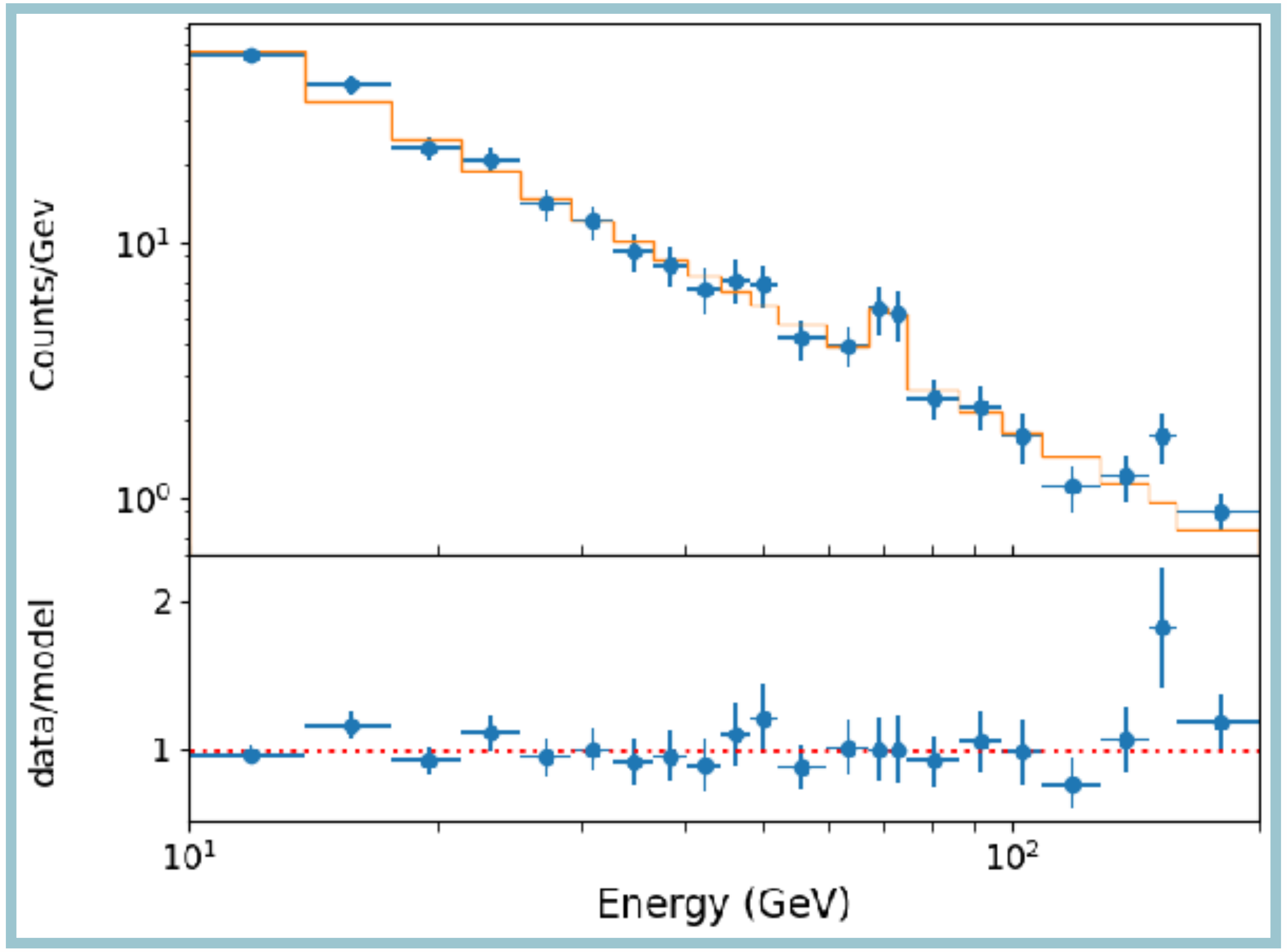


Have other people calculated the distribution of colours? What can we reason about p(f) ?

Assumption that all colours are mixed together at the factory and then distributed into packages: binomial likelihood

What's the probability of a certain fraction of blue smarties given our observations, p(f | b) ?

What about physics?



Data: γ-ray spectrum

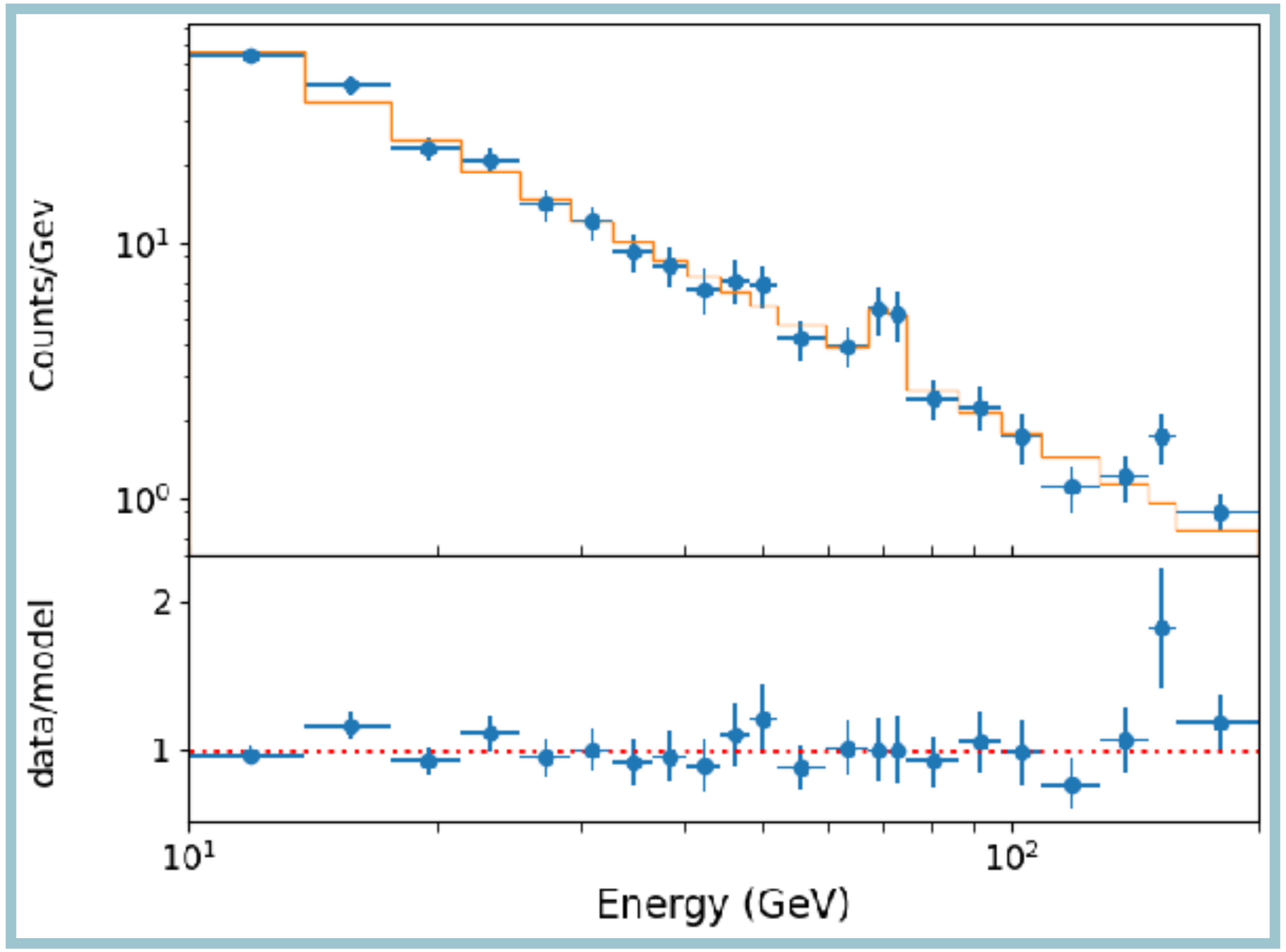
Model: Power law + Gaussian line

Likelihood: Poisson

Parameters: A_{PL}, Γ, A_{Line}, E_{Line}, $\sigma_{\mathrm{Line}} A_{\mathrm{Line}}$

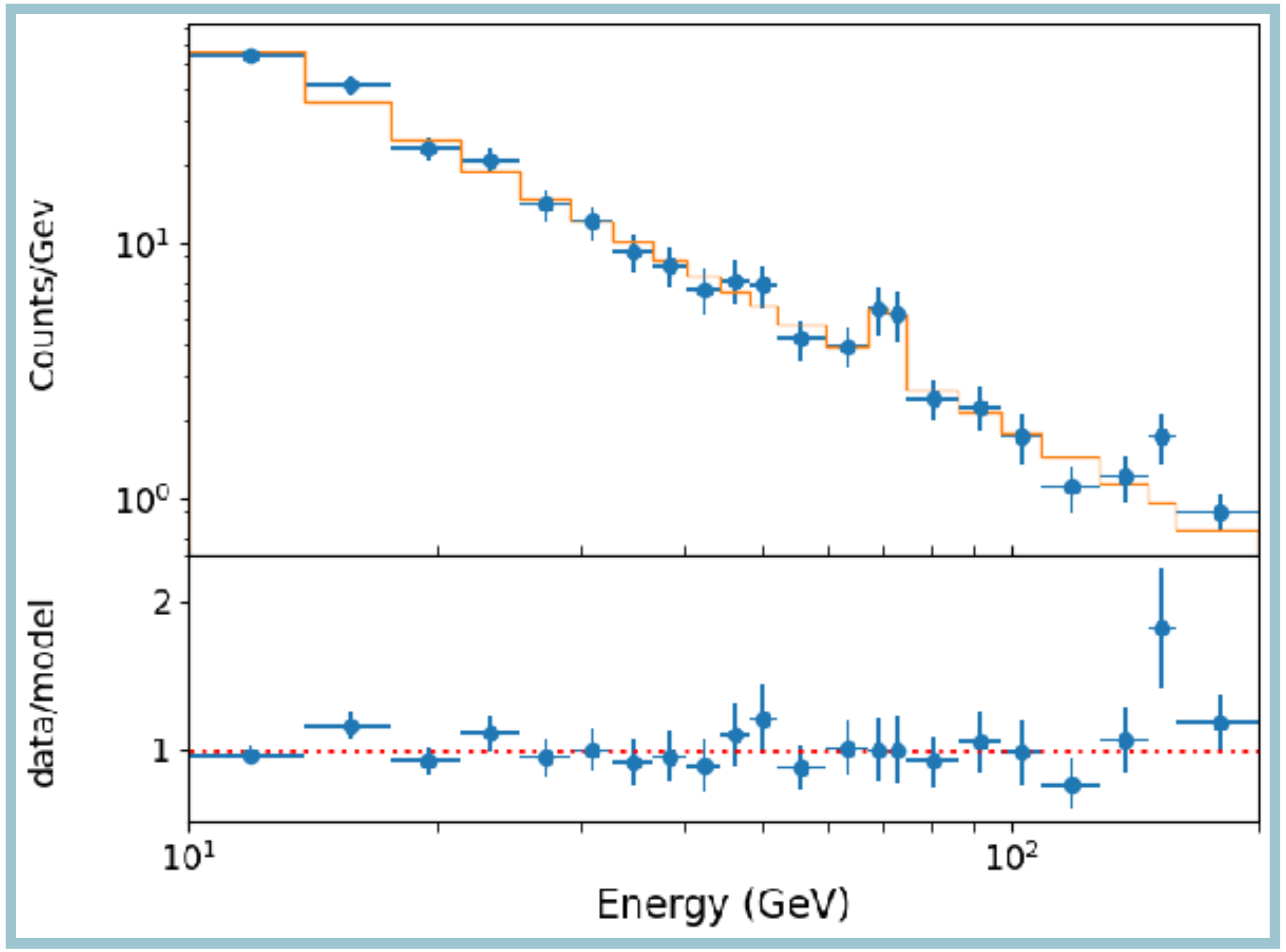
Priors: ???

What about physics?



Data:	**X-ray spectrum**
Model:	**Power law + Gaussian line**
Likelihood:	**Poisson**
Parameters:	$A_{\mathrm{PL}}, \Gamma, A_{\mathrm{Line}}, E_{\mathrm{Line}}, \sigma_{\mathrm{Line}} A_{\mathrm{Line}}$
Priors:	**???**

What about physics?



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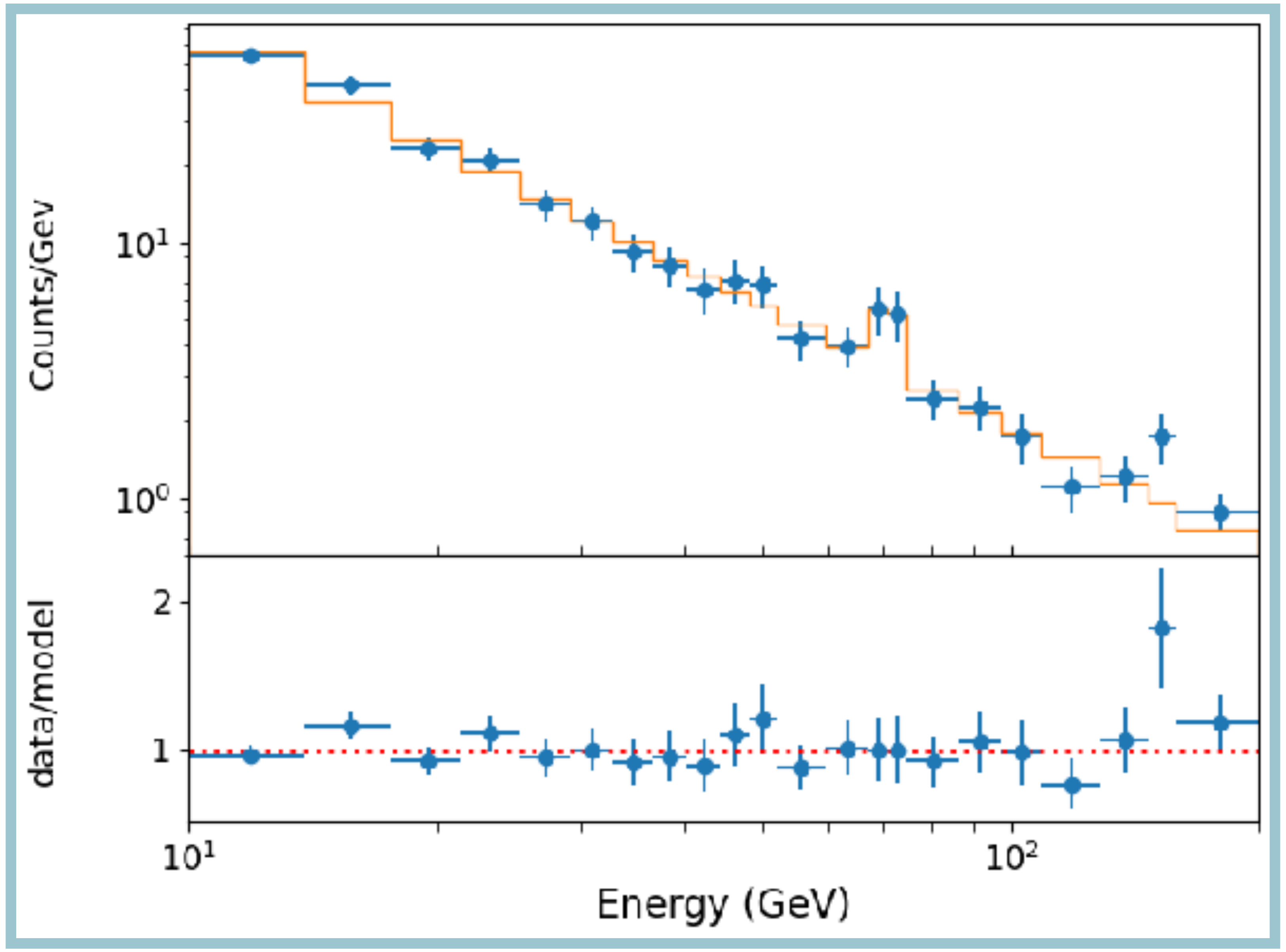
Parameters: $A_{\mathrm{PL}}, \Gamma, A_{\mathrm{Line}}, E_{\mathrm{Line}}, \sigma_{\mathrm{Line}} A_{\mathrm{Line}}$

Priors: ???

$$p(A_{\mathrm{PL}}, \Gamma, A_{\mathrm{Line}}, E_{\mathrm{Line}}, \sigma_{\mathrm{Line}} \,|\, D) = \frac{p(D \,|\, A_{\mathrm{PL}}, \Gamma, A_{\mathrm{Line}}, E_{\mathrm{Line}}, \sigma_{\mathrm{Line}}) p(A_{\mathrm{PL}}, \Gamma, A_{\mathrm{Line}}, E_{\mathrm{Line}}, \sigma_{\mathrm{Line}})}{p(D)}$$

What about physics?



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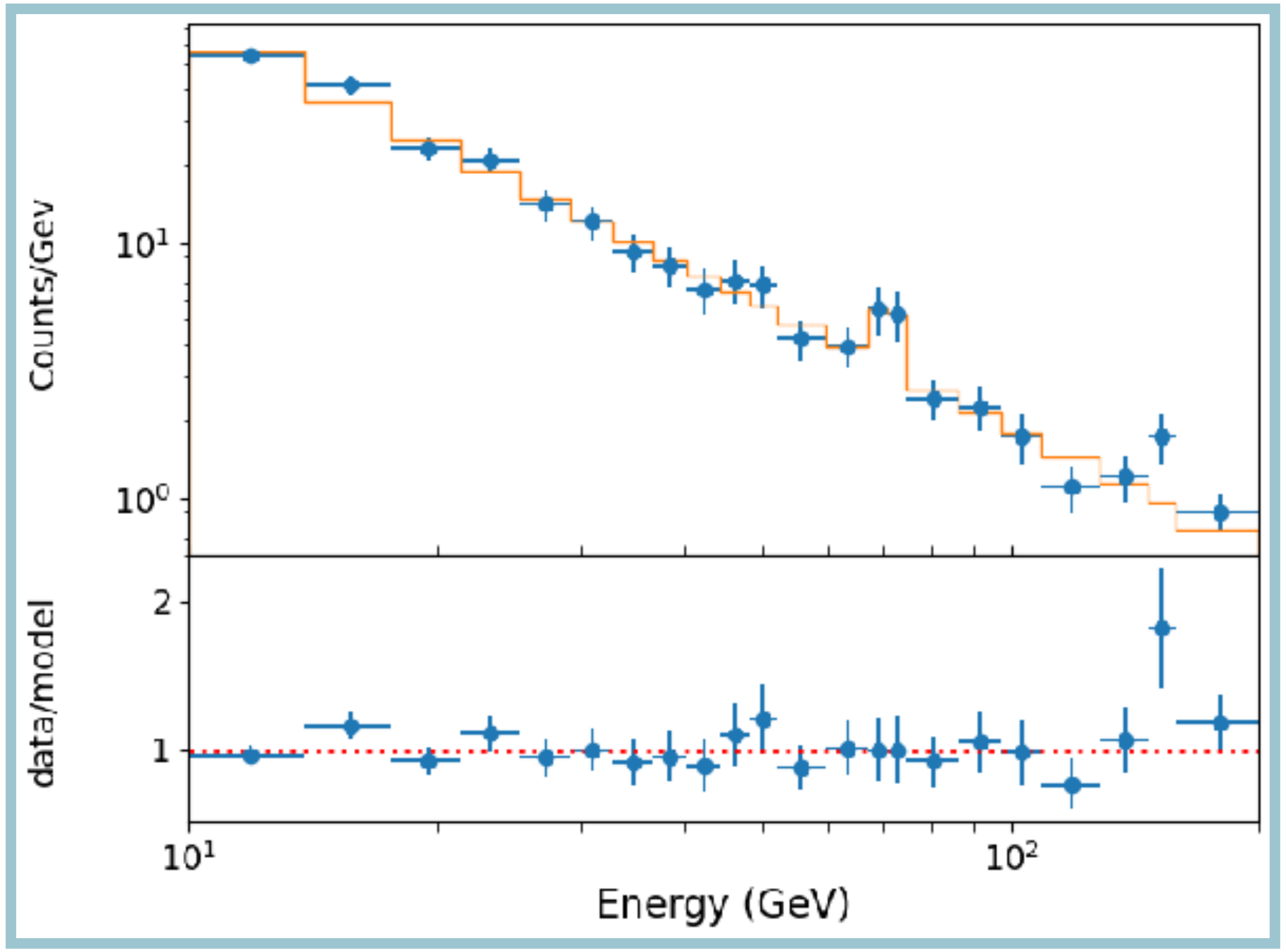

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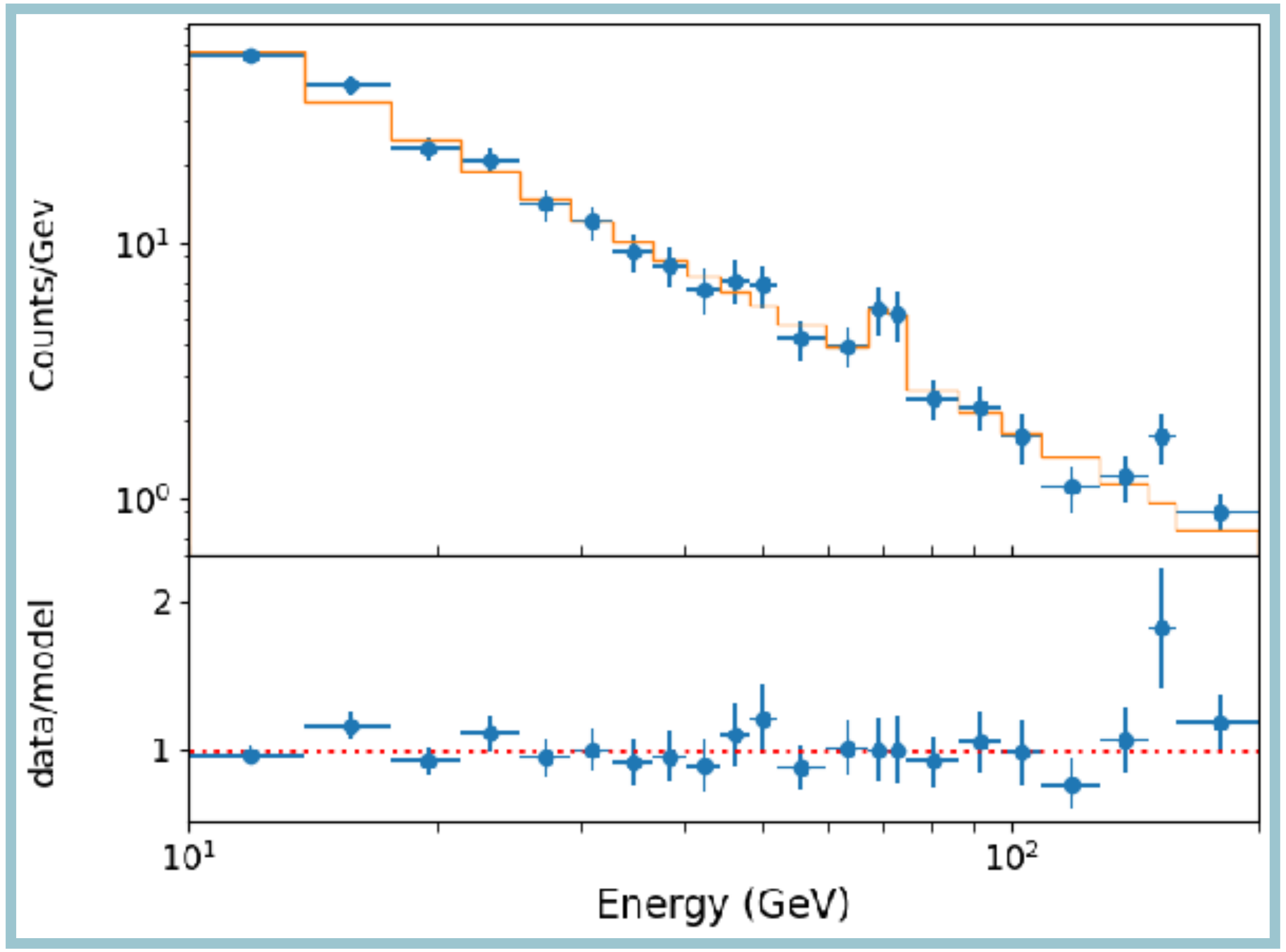
What assumption have we made here?

What about physics?



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$$p(A_{\mathrm{PL}}, \Gamma, A_{\mathrm{Line}}, E_{\mathrm{Line}}, \sigma_{\mathrm{Line}} \,|\, D) = \frac{p(D \,|\, A_{\mathrm{PL}}, \Gamma, A_{\mathrm{Line}}, E_{\mathrm{Line}}, \sigma_{\mathrm{Line}}) p(A_{\mathrm{PL}}, \Gamma, A_{\mathrm{Line}}, E_{\mathrm{Line}}, \sigma_{\mathrm{Line}})}{p(D)}$$

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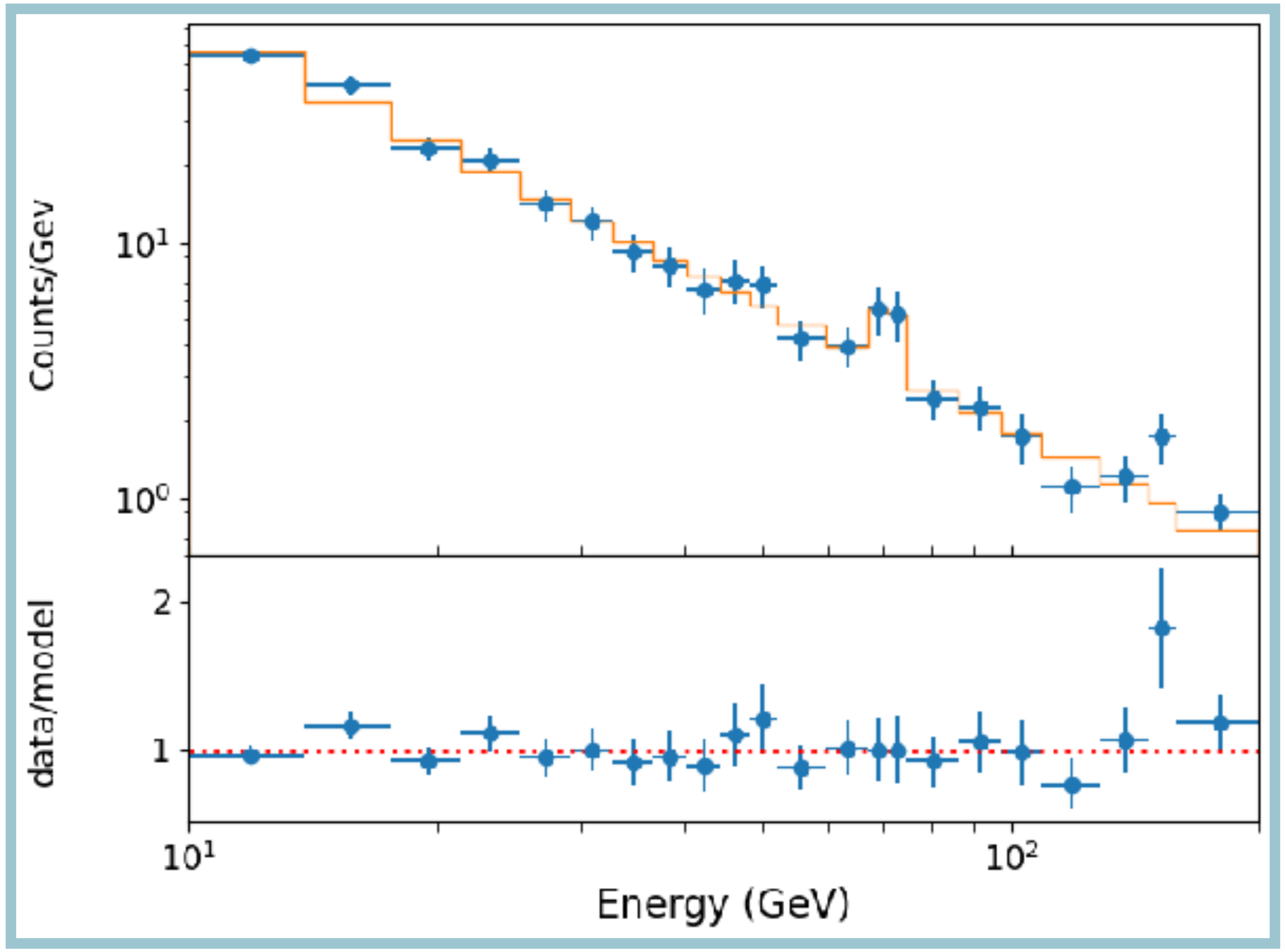
What assumption have we made here?

Independence of priors!

What about physics?



Data: **X-ray spectrum**

Model: **Power law + Gaussian line**

Likelihood: **Poisson**

Parameters: $A_{\mathrm{PL}}, \Gamma, A_{\mathrm{Line}}, E_{\mathrm{Line}}, \sigma_{\mathrm{Line}} A_{\mathrm{Line}}$

Priors: **???**

What priors should we choose?

$$p(A_{\mathrm{PL}}, \Gamma, A_{\mathrm{Line}}, E_{\mathrm{Line}}, \sigma_{\mathrm{Line}} \,|\, D) = \frac{p(D \,|\, A_{\mathrm{PL}}, \Gamma, A_{\mathrm{Line}}, E_{\mathrm{Line}}, \sigma_{\mathrm{Line}}) p(A_{\mathrm{PL}}, \Gamma, A_{\mathrm{Line}}, E_{\mathrm{Line}}, \sigma_{\mathrm{Line}})}{p(D)}$$

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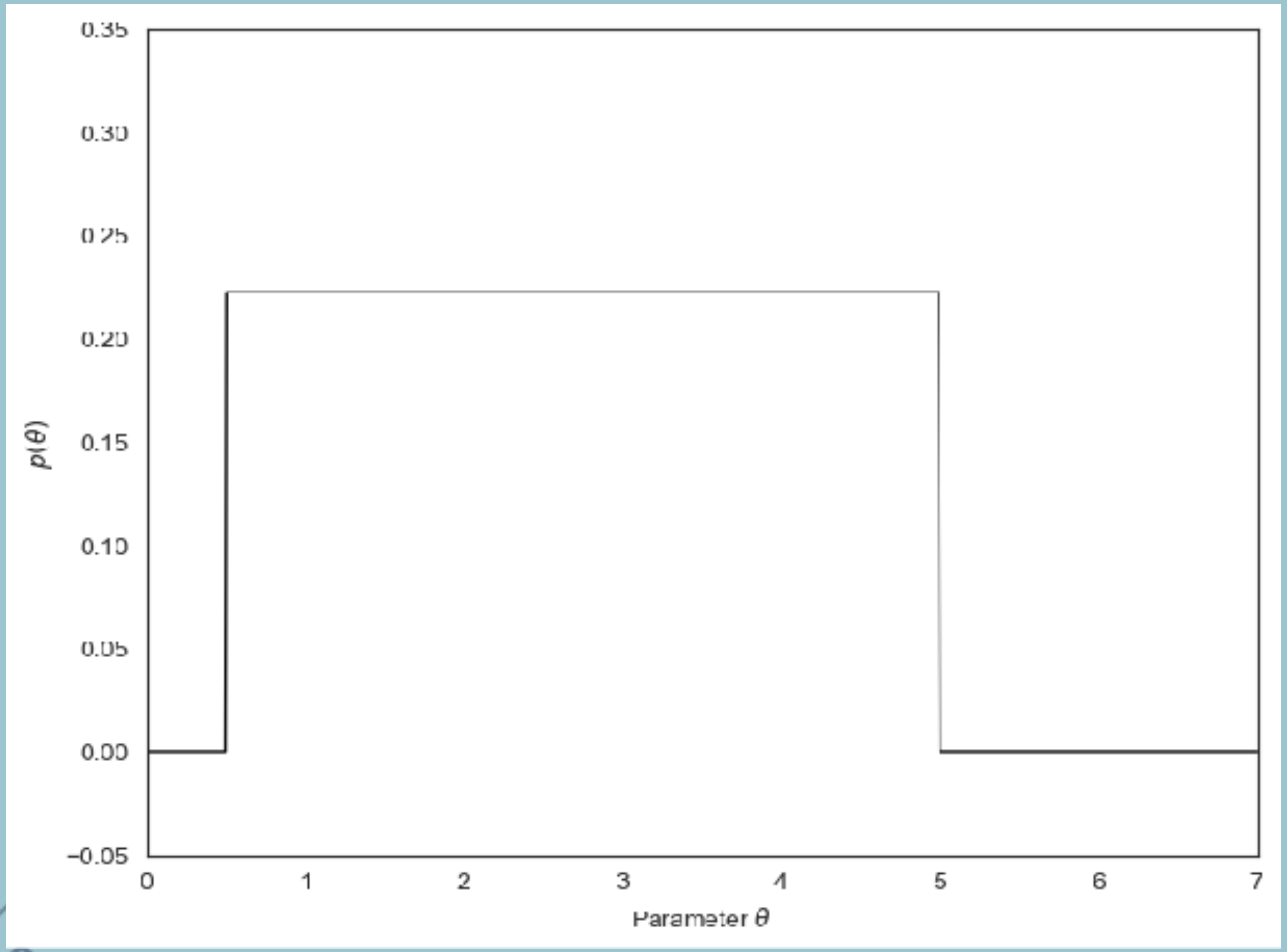
What assumption have we made here?

Independence of priors!

Flat Priors:
Γ

"I know nothing but I'm 100% sure that my parameter is between $\theta_{\min}$ and $\theta_{\max}$



Scale-free priors:

A_{PL}, A_{Line}

"I believe with equal prior probability that my parameter could be between 0.1 and 1 or between 1 and 10"

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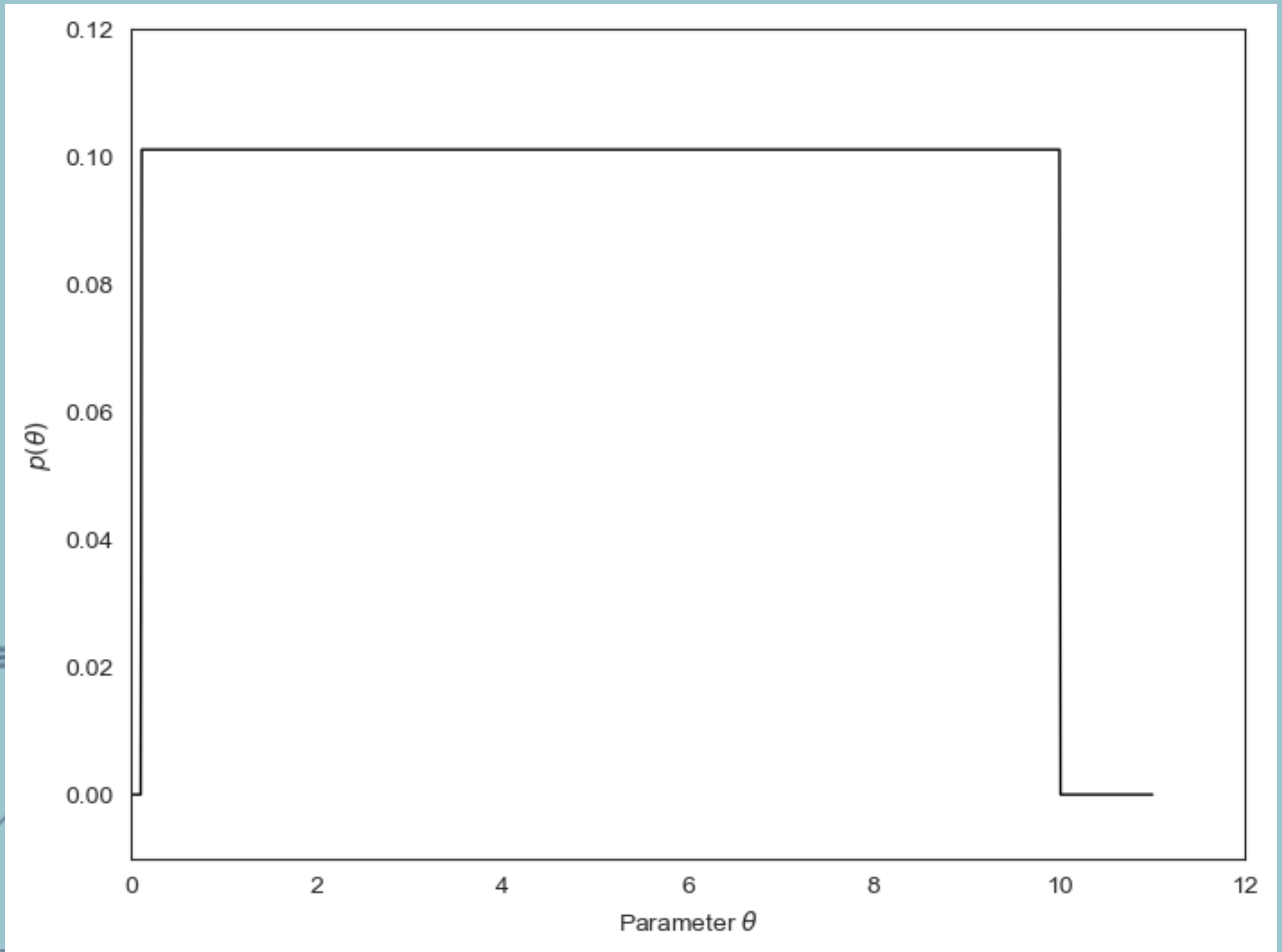

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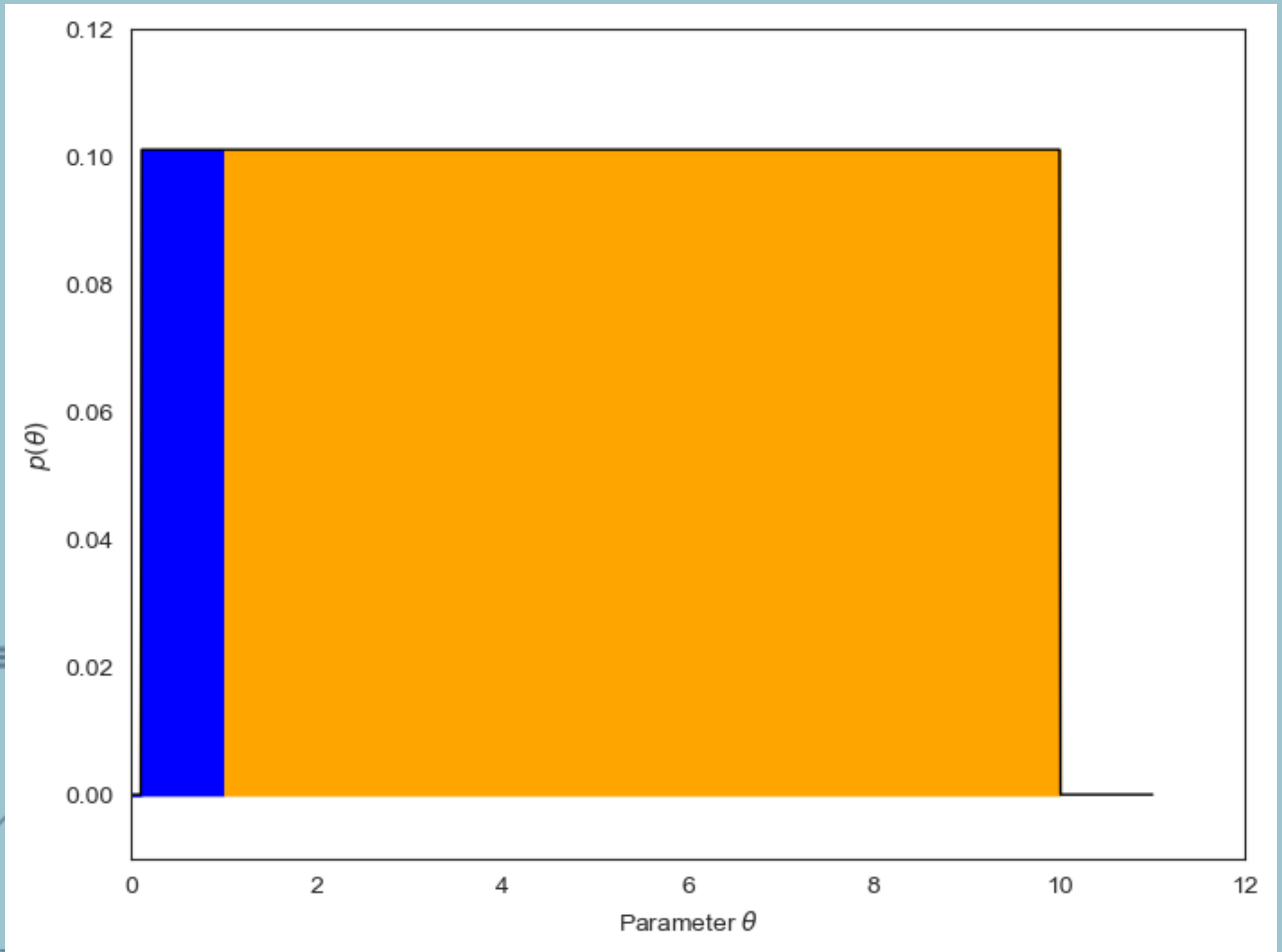


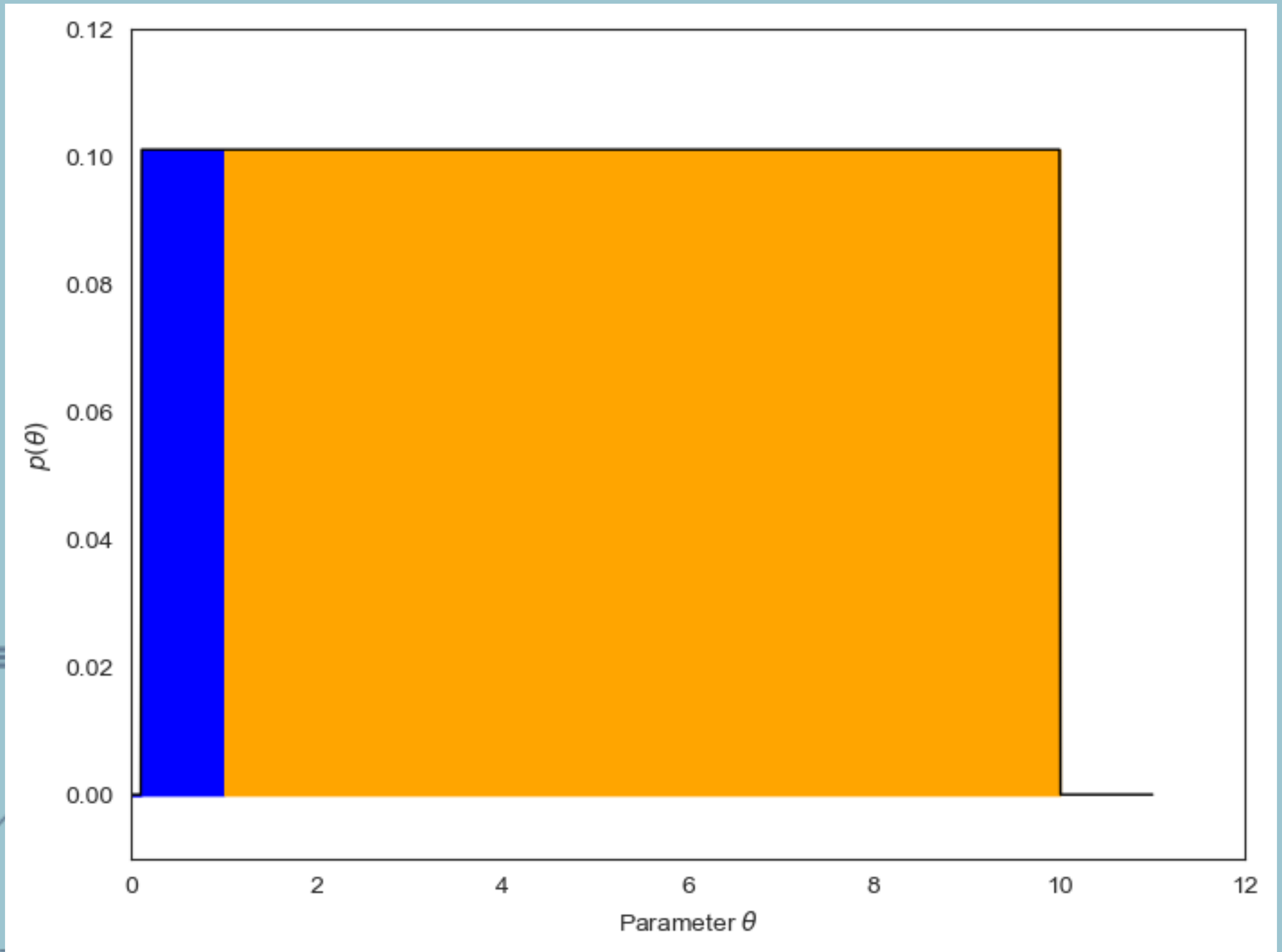
Scale-free priors:

A_{PL}, A_{Line}

$\downarrow$

$\log(A_{PL})$, $\log(A_{Line})$

"I believe with equal prior probability that my parameter could be between 0.1 and 1 or between 1 and 10"



Scale-free priors:

$A_{\mathrm{PL}}, A_{\mathrm{Line}}$

$\downarrow$

$\log(A_{\mathrm{PL}}), \log(A_{\mathrm{Line}})$

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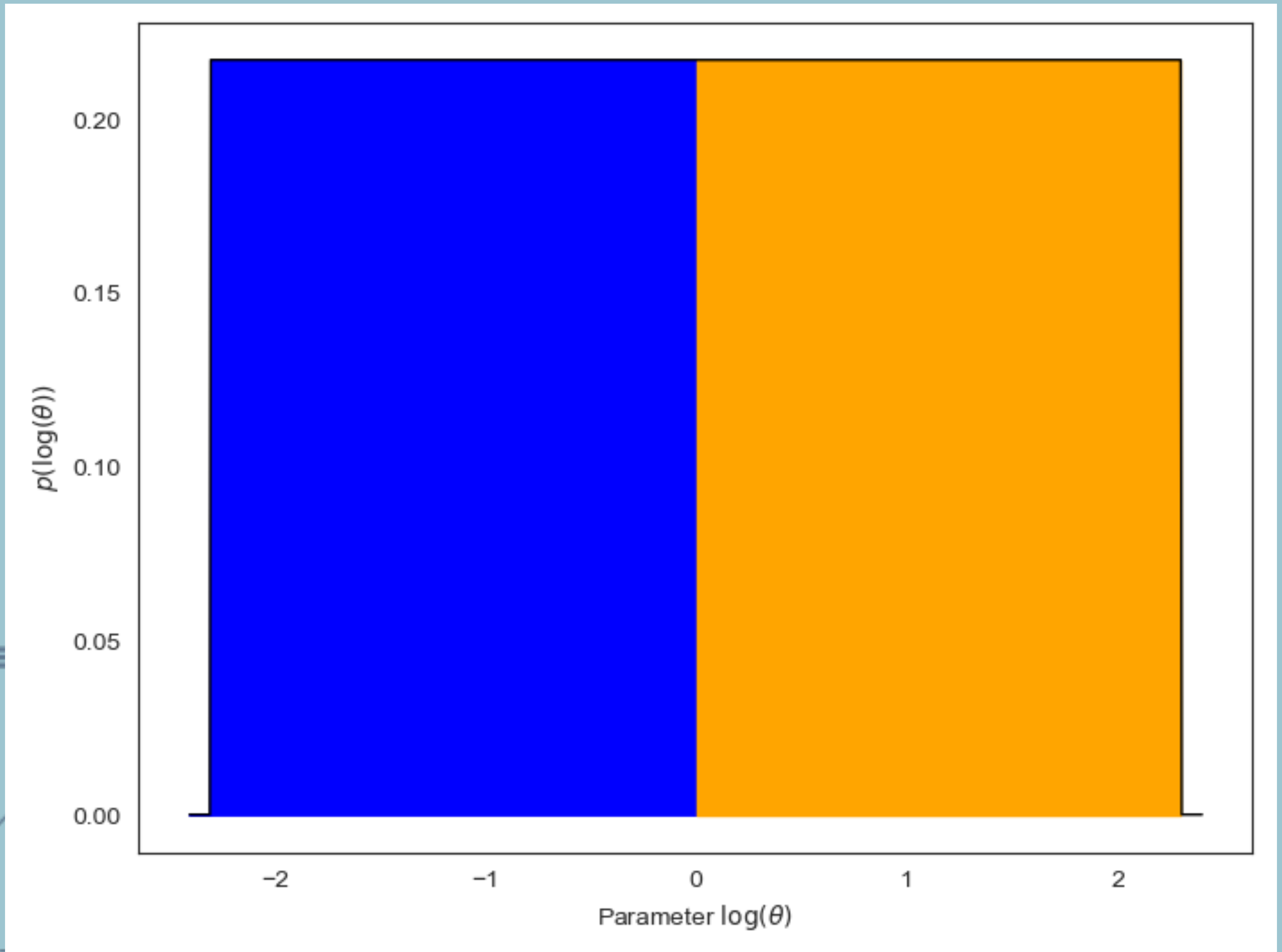


Scale-free priors:

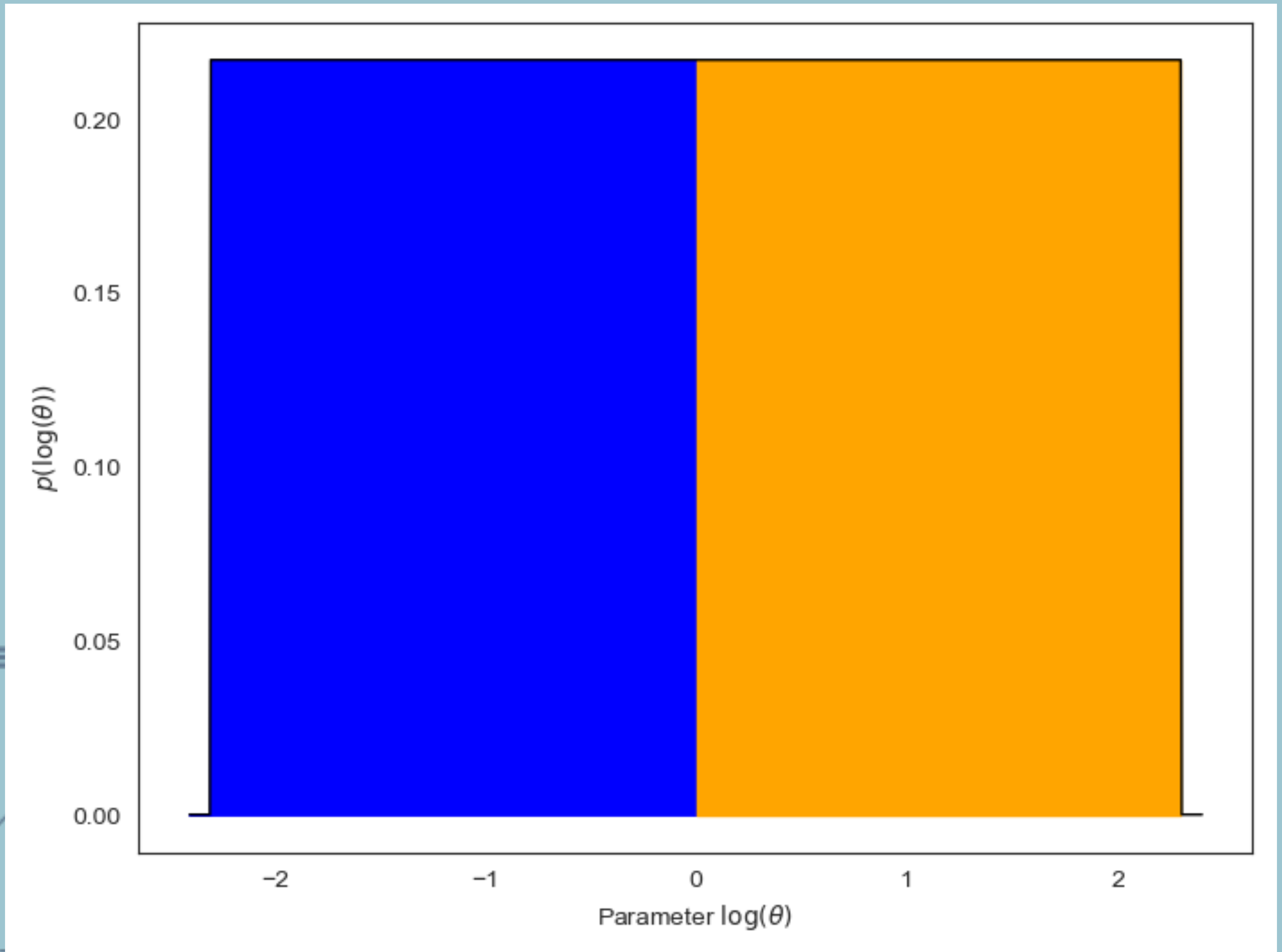
A_{PL}, A_{Line}

$\downarrow$

$\log(A_{\mathrm{PL}})$, $\log(A_{\mathrm{Line}})$

Also use this for parameters that cannot be negative!

"I believe with equal prior probability that my parameter could be between 0.1 and 1 or between 1 and 10"

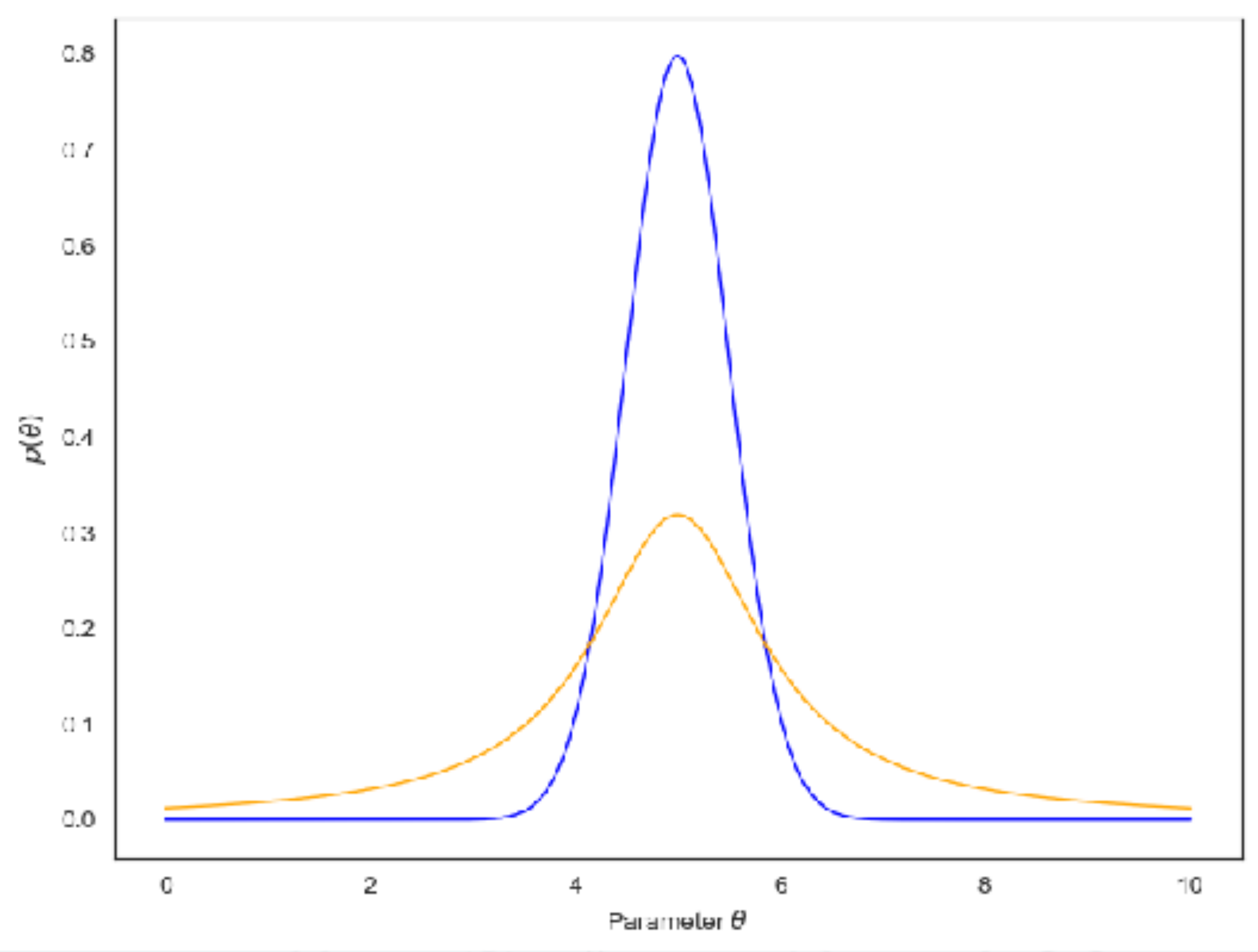


Other Distributions:

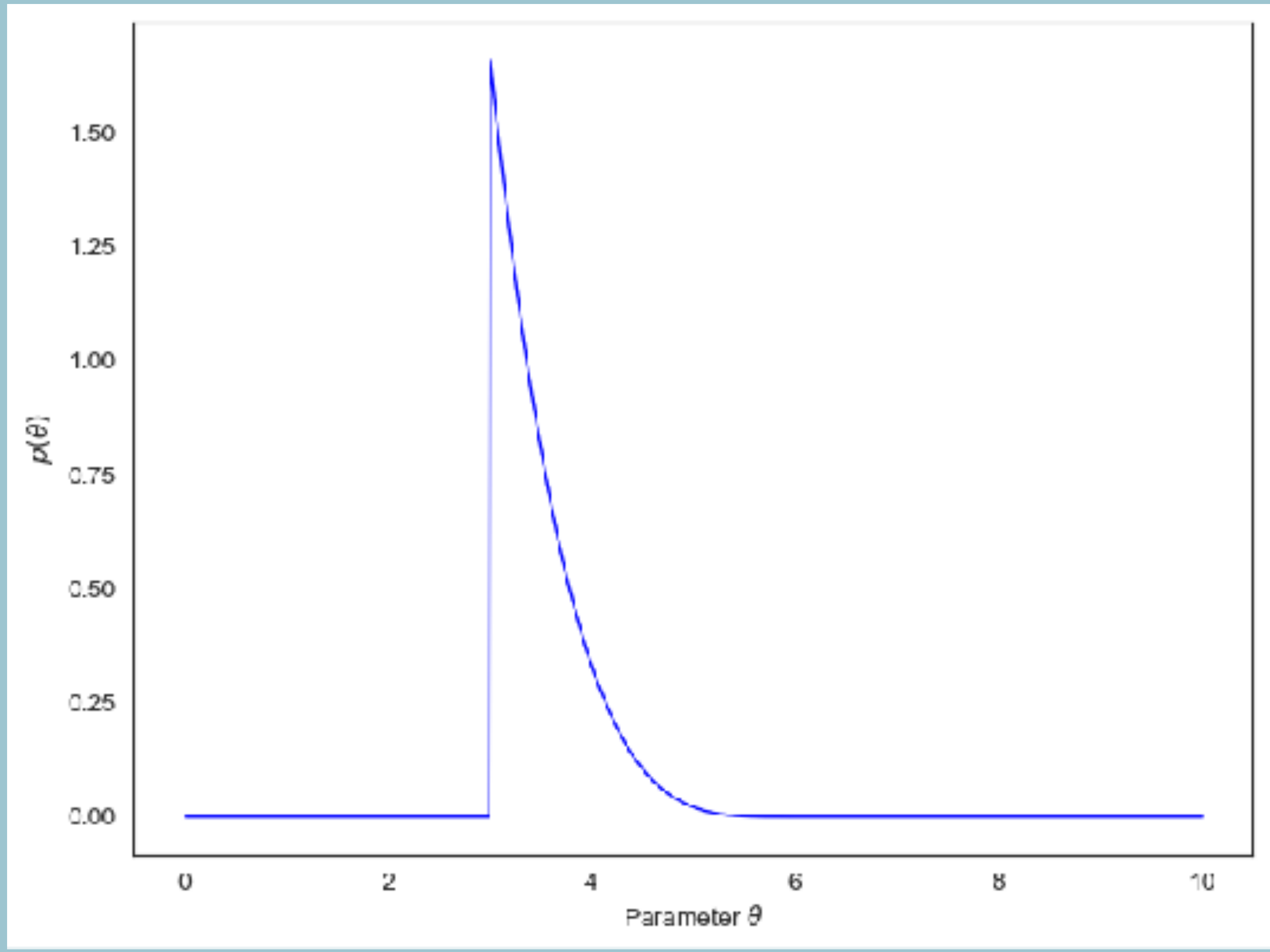
E_{line}

"I have prior information that says that my parameter θ should be around a value of $\hat{\theta}$, but I'm not sure."

Gaussian and t-distribution



Beta-distribution



Reparametrizations

σ_{line}

"The possible width of a line generally scales with the line's energy."

$$p(E_{\text{line}}, \sigma_{\text{line}}) \neq p(E_{\text{line}})p(\sigma_{\text{line}})$$

Reparametrizations

σ_{line}

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$$\downarrow$$

$$q = \frac{E_{\text{line}}}{\sigma_{\text{line}}}$$

Reparametrizations

σ_{line}

"The possible width of a line generally scales with the line's energy."

$$p(E_{\text{line}}, \sigma_{\text{line}}) \neq p(E_{\text{line}})p(\sigma_{\text{line}})$$

$$\downarrow$$

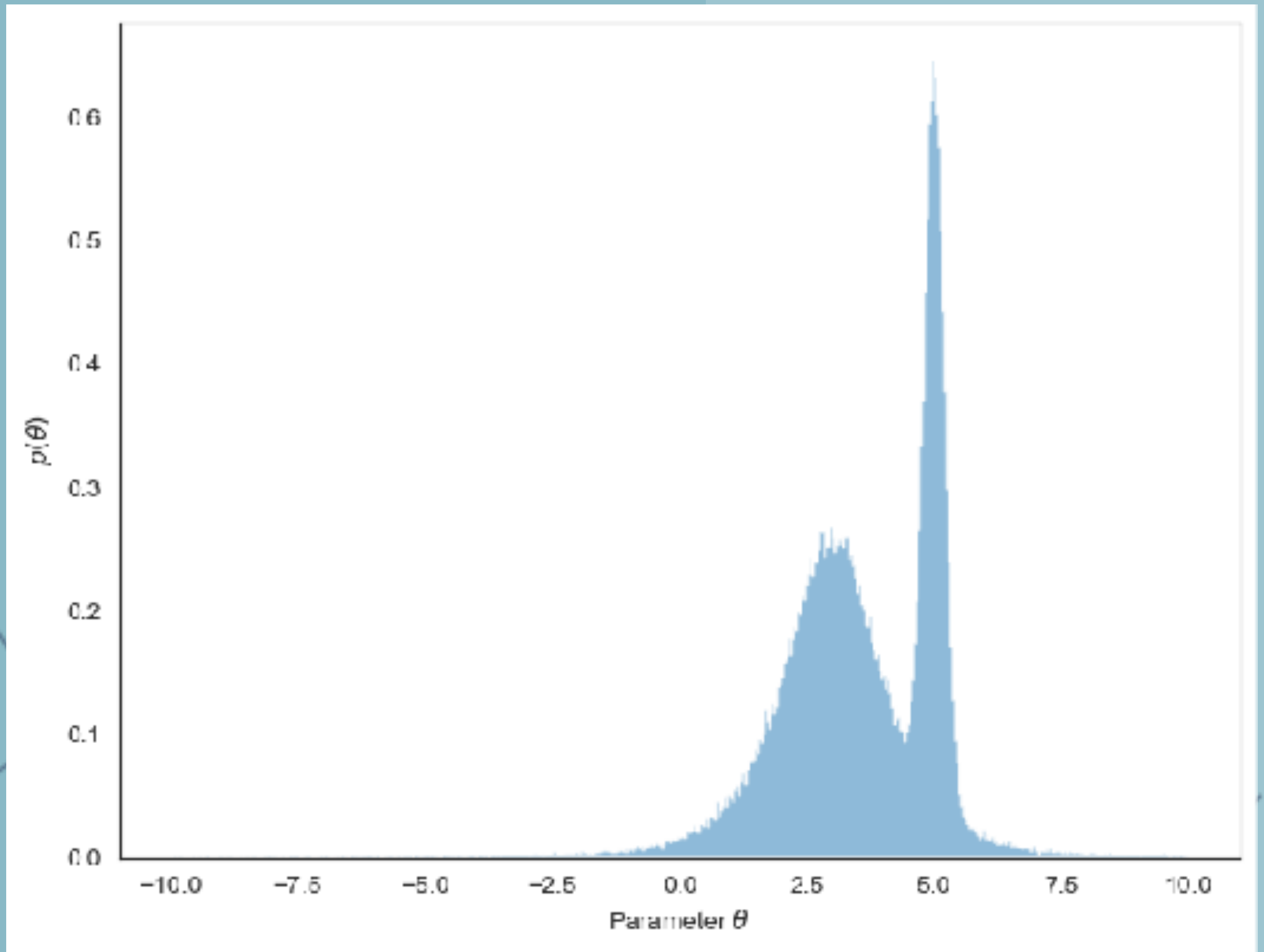
$$q = \frac{E_{\text{line}}}{\sigma_{\text{line}}}$$

$$\downarrow$$

$$p(E_{\text{line}}, q) = p(E_{\text{line}})p(q)$$

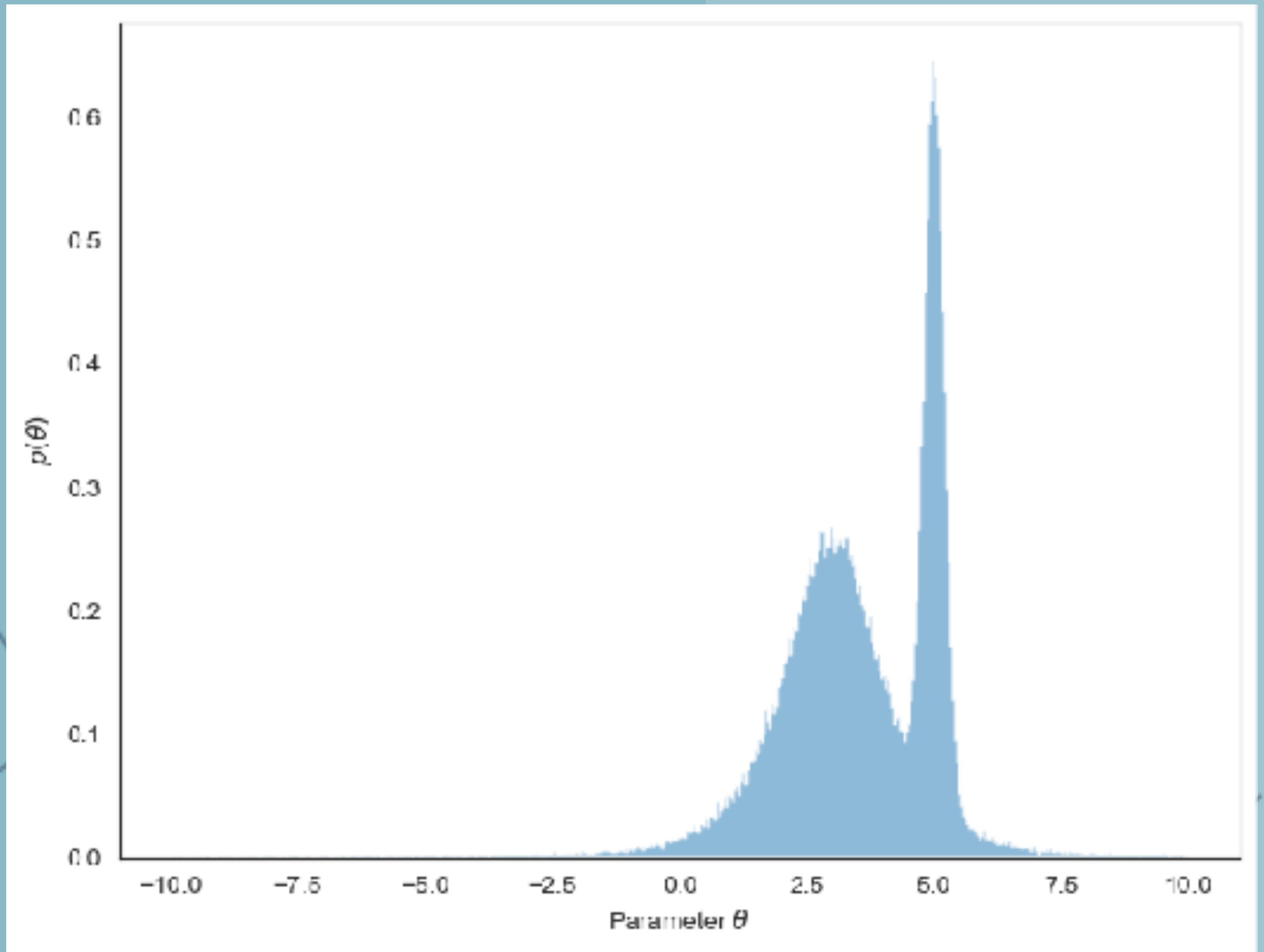
Empirical Distributions

"My prior is actually someone else's posterior"

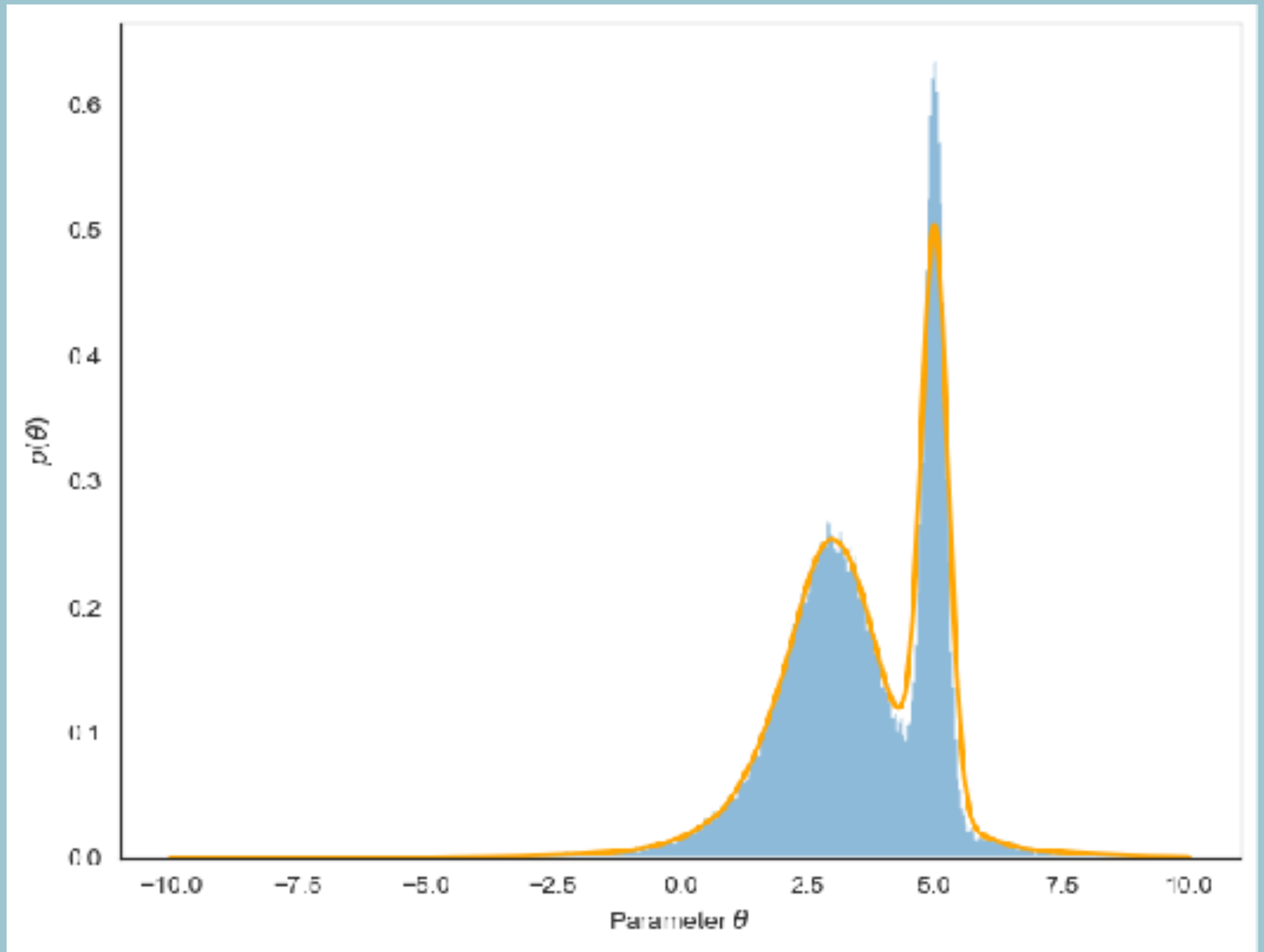


Empirical Distributions

“My prior is actually someone else’s posterior”

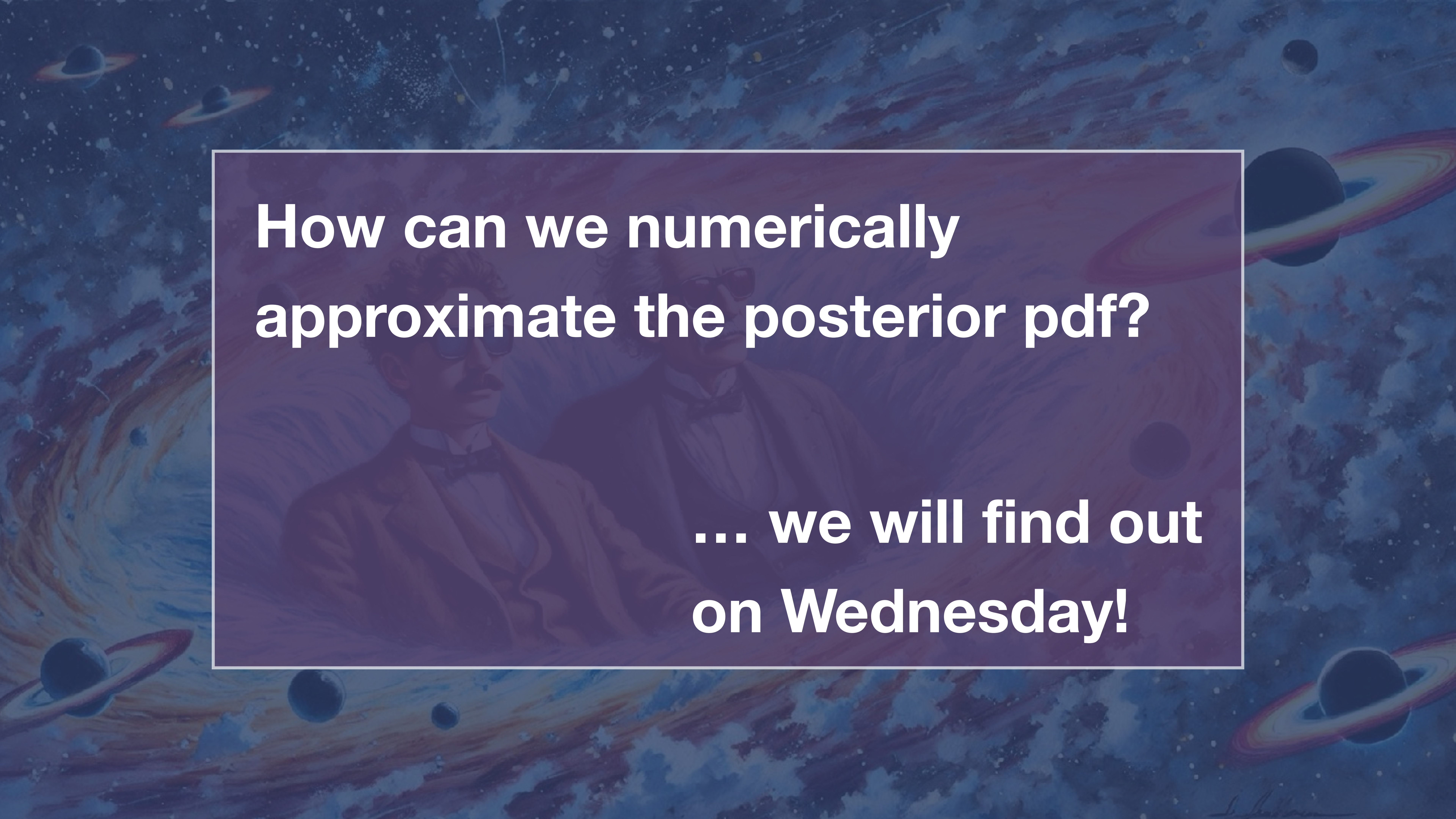


Kernel Density Estimation



We’ve chosen priors, but they’re not conjugate! 😱

How can we numerically approximate the posterior pdf?


How can we numerically
approximate the posterior pdf?
… we will find out
on Wednesday!